Special Soundness in the Random Oracle Model

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Abstract. We generalize the knowledge extractor for constant-round special sound protocols presented by Wikström (2018) to a knowledge extractor for the corresponding non-interactive Fiat-Shamir proofs in the random oracle model and give an exact analysis of the extraction error and running time.
Relative the interactive case the extraction error is increased by a factor ℓ and the running time is increased by a factor $O(\ell)$, where ℓ is the number of oracle queries made by the prover.
Through carefully chosen notation and concepts, and a technical lemma, we effectively recast the extraction problem of the notoriously complex non-interactive case to the interactive case. Thus, our approach may be of independent interest.

1 Introduction

Zero knowledge proofs and proofs of knowledge. Zero knowledge proofs were discovered by Goldwasser, Micali, and Rackoff [10]. They allow a prover to interactively convince a verifier that a statement is true without disclosing anything else. A related notion discovered by Bellare and Goldreich [5], are proofs of *knowledge*. In such protocols the prover not only shows that a statement is true, but that it holds a witness of this fact.

The completeness of a protocol is the probability that it completes successfully when both parties follow the protocol on a valid common input. The *soundness error* of a protocol is the probability that a malicious prover convinces an honest verifier that a false statement is true.

If there is an extraction algorithm such that for every prover and every statement a witness is output in expected time (over the internal randomness of the extractor) $\mathsf{poly}/(\Delta - \epsilon)$, where Δ is the probability that the honest verifier is convinced and ϵ is the *knowledge error*, then the protocol is called a proof of knowledge. Thus, the knowledge error is an upper bound on the probability that a prover convinces a verifier without knowing a witness.

The extractor may rewind and complete multiple executions from any point of the execution, i.e., it treats the prover as a deterministic oracle. A knowledge error ϵ implies a soundness error of at most ϵ, since the analysis of the knowledge extractor may be seen as a probabilistic proof [1]. Due to the efficiency requirement on the extractor the reverse implication does not hold. Readers are referred to [9] for a thorough discussion of variations of these notions.

随机预言模型中的特殊可靠性

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摘要。我们将Wikström（2018年）提出的常轮次特殊可靠性协议的知识提取器推广到随机预言模型中相应的非交互式Fiat-Shamir证明的知识提取器，并对提取错误和运行时间进行了精确分析。相对于交互式情况，提取错误增加了 ℓ 倍，运行时间增加了 $O(\ell)$倍，其中 ℓ 是证明者发出的预言机查询次数。通过精心选择的符号和概念，以及一个技术引理，我们将这个以复杂著称的非交互式情况提取问题有效地转化为交互式情况。因此，我们的方法可能具有独立的研究价值。

1 引言

零知识证明和知识证明。零知识证明由Goldwasser、Micali和Rackoff [10]发现。它们允许证明者交互式地让验证者相信一个陈述为真，而不透露任何其他信息。Bellare和Goldreich [5], 发现的一个相关概念是知识证明。在这样的协议中，证明者不仅表明一个陈述为真，而且表明自己持有该事实的见证。

协议的完备性是指当双方遵循协议在有效的共同输入上时，协议成功完成的概率。协议的可信度错误是指恶意证明者让诚实验证者相信错误陈述为真的概率。

如果存在一个提取算法，使得对于每个证明者和每个陈述，见证在期望时间内（在提取器的内部随机性上）$\mathsf{poly}/(\Delta - \epsilon)$输出，其中 Δ 是诚实验证者被说服的概率， ϵ 是知识错误，那么该协议被称为知识证明。因此，知识错误是证明者在不了解见证的情况下让验证者相信陈述为真的概率的上界。

提取器可以从执行的任何一点回绕并完成多次执行，即，它将证明者视为一个确定性预言机。知识错误 ϵ 意味着一个至多 ϵ的可靠性错误，因为知识提取器的分析可以被视为一个概率证明 [1]。由于对提取器效率的要求，反向蕴涵不成立。读者可参考 [9] 以深入了解这些概念的各种变体。

Special soundness. A three-message public-coin protocol [10,3] is defined to be *special sound* if a witness can be computed efficiently from two accepting transcripts with a common first prover message, but distinct verifier messages. This notion was introduced by Cramer et al. [7] as a generalization of a property of Schnorr's proof of knowledge of a discrete logarithm [11].

In the generalization of Wikström [12] a $(2r+1)$-round protocol is special sound if: (1) the ith verifier message is chosen uniformly at random from the ground set S_i of a matroid $\mathbb{M}_i$ for $i \in [r]$, and (2) a witness can be computed from a tree of accepting transcripts such that for each $i \in [r]$ and each node at depth $i-1$ the verifier messages form a basis of $\mathbb{M}_i$. Matroids capture independence abstractly, but examples from the literature include inequality [11] or linear independence [4].

Fiat-Shamir heuristic. Recall that a public coin protocol may be converted into a non-interactive protocol using the Fiat-Shamir transform [8]. This replaces each verifier message by the output of a hash function evaluated on the common input and the current partial transcript. This is important in practice to reduce the number of rounds.

The Fiat-Shamir heuristic suggests that we may analyze such protocols by replacing the hash function by a random oracle. The random oracle model was generalized and formalized by Bellare and Rogaway [6]. When we analyze the protocol in the random oracle model we are effectively assuming that it suffices to consider adversaries which treats the hash function as if it was ideal and never inspects its definition.

However, the adversary may still exploit the fact that it may query the random oracle repeatedly on inputs of its choice. This enables it to probe a tree of partial executions until it can extend at least one interaction with the random oracle to a complete accepting transcript that it outputs as its non-interactive proof.

Grafting protocols. It is more convenient to think of the interaction between the prover and the random oracle as a *grafting protocol* where the prover may extend the execution from any previous existing verifier message by grafting a new branch, i.e., a reply, to a verifier message. The verifier is easily adapted correspondly to give the prover this ability. An execution is then considered to be accepting if any path from the root to a leaf in the resulting tree of transcripts, corresponding to a transcript of the basic protocol, is accepting.

We stress that neither party rewinds to a previous point in the execution; branches are *grafted* to the existing tree of executions which remains part of the view. Furthermore, the branches are added in a particular order by the prover, i.e., each prover-verifier message exchange is associated with an integer index. This means that an execution of the grafting protocol may be identified with a topologically ordered subtree of the tree of all possible executions of the basic protocol.

Provided that the verifier messages have high entropy the computation of a Fiat-Shamir proof is statistically close in distribution to the execution of the corresponding grafting protocol. Thus, we study knowledge extraction for grafting protocols.

1.1 Contribution

It is well known [11] that if a prover convinces a verifier with probability Δ in a three-message special sound protocol, then a witness can be extracted in expected time

特殊安全性。一个三消息公开硬币协议 [10,3] 被定义为特殊安全，如果可以从两个接受性转录中高效地计算出一个证明者，这两个转录具有相同的第一条证明者消息，但验证者消息不同。这个概念由 Cramer 等人 [7] 引入，作为 Schnorr 离散对数知识证明 [11] 的一个性质的推广。

在Wikström [12] a $(2r+1)$-轮协议的泛化中，如果满足以下条件，则该协议是特殊的：1) 第 i 个验证器消息是从 matroid $\mathbb{M}_i$ 的基集 S_i 中均匀随机选择的，且该 matroid 是针对 $i \in [r]$ 的；2) 可以从接受转录树的树中计算出一个证据，使得对于每个 $i \in [r]$ 和每个深度为 $i-1$ 的节点，验证器消息构成 $\mathbb{M}_i$ 的一个基。Matroid 抽象地捕捉独立性，但文献中的例子包括不等式 [11] 或线性无关性 [4]。

Fiat-Shamir 启发式。回想一下，可以使用 Fiat-Shamir 变换 [8] 将公开硬币协议转换为非交互式协议。该变换用关于公共输入和当前部分转录的哈希函数的输出替换每个验证器消息。这在实践中很重要，可以减少轮数。

Fiat-Shamir 启发式表明，我们可以通过用随机预言机替换哈希函数来分析此类协议。随机预言机模型由 Bellare 和 Rogaway [6] 进行了泛化和形式化。当我们在随机预言机模型中分析协议时，我们实际上是在假设，考虑的对手将哈希函数视为理想的，并且永远不会检查其定义，这足以进行分析。

然而，攻击者仍然可以利用它可以选择性地多次查询随机预言机的输入这一事实。这使其能够探测部分执行树，直到它可以将至少一个与随机预言机的交互扩展为完整的接受式会话记录，并将其作为其非交互式证明输出。

嫁接协议。将证明者与随机预言机之间的交互视为嫁接协议更为方便，其中证明者可以通过将一个新分支（即回复）嫁接到验证者消息上，来扩展从任何先前存在的验证者消息开始的执行。验证者相应地很容易进行调整，以赋予证明者这种能力。然后，如果结果会话记录树中从根到叶子的任何路径（对应于基本协议的会话记录）是接受的，则执行被视为接受。

我们强调，双方都不会回滚到执行的先前点；分支被嫁接到现有的执行树上，该树仍然是视图的一部分。此外，证明者按特定顺序添加分支，即每个证明者-验证者消息交换都与一个整数索引相关联。这意味着嫁接协议的执行可以与基本协议所有可能执行树中的拓扑有序子树相对应。

如果验证器消息具有高熵，那么Fiat-Shamir证明的计算在统计分布上与相应嫁接协议的执行非常接近。因此，我们研究嫁接协议的知识提取。

1.1 贡献

众所周知，如果证明者在一个三消息特殊安全协议中用概率 [11] 使验证者信服，那么就可以在期望时间内提取出证物

$p/(\Delta - \epsilon)$, where p is polynomial and ϵ is the knowledge error. Wikström [12] generalized the notion to constant-round special sound protocols and gave an exact and tight bound. The concrete contribution of this work is a corresponding theorem for grafting protocols.

Theorem 1 (Informal). *Let $(\mathcal{P}, \mathcal{V})$ be a $(2r+1)$-message $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$-special sound protocol with soundness error ϵ_S and knowledge error ϵ_K for a knowledge extractor that for any instance and prover that convinces the verifier with probability $\Delta > \epsilon_K$, for a constant c is expected to execute the protocol $c/(\Delta - \epsilon_K)$ times.*

Then its $(2\ell + 1)$-message grafting protocol $(\mathsf{G}[\mathcal{P}], \mathsf{G}[\mathcal{V}])$ has soundness error $\ell\epsilon_S$ and knowledge error $\ell\epsilon_K$ for a knowledge extractor that for any instance and prover that convinces the verifier with probability $\Delta > \ell\epsilon_K$ is expected to execute the grafting protocol $4 \cdot 3^{r+1}\ell \cdot c/(\Delta - \ell\epsilon_K)$ times.

In applications we may often choose parameters of the protocol to reduce the soundness and knowledge errors by a factor of $1/\ell$. Thus, in practice the Fiat-Shamir transform causes a loss of roughly $\log \ell + O(1)$ bits of security.

A careful choice of notation and concepts, and a technical lemma, allow us to effectively reduce the problem of constructing an extractor to the combinatorial problem of finding an accepting basis in a suitably defined matroid tree, but the distribution of the verifier messages is influenced by the adversary and not necessarily uniform. The main technical challenge is to prove that this distribution can be sampled efficiently. Apart from this the analysis from [12] applies mutatis mutandi.

Remark. Attema, Fehr, and Klooss recently informed us, that a few months after our discovery, they independently discovered a theorem similar to our main theorem. Interestingly, their work is based on Attema et al. [2], while we rely on the work of Wikström [12]. Thus, in future work we hope to understand the strengths and weaknesses of both approaches.

1.2 Proof Strategy

Grafting protocols. We first formalize the computation of a non-interactive Fiat-Shamir proof, based on a special sound protocol, as the execution of a *grafting protocol*. In such protocols the verifier allows the prover to repeatedly: (1) spawn a new execution of the basic protocol, or (2) extend an existing partial execution of the basic protocol by grafting an additional round to it.

Thus, at any point during execution the transcript may be viewed as a tree of partial executions that grows one edge for each round. We stress that the prover may choose the location of each grafted round and that this may depend on both the structure of the tree and the verifier messages seen so far. The verifier accepts if there exists an embedded accepting transcript of the basic protocol corresponding to a path from the root to a leaf.

This captures the computation of a Fiat-Shamir proof faithfully except that the prover may only execute a round after the previous rounds have been executed, i.e., it is effectively restricted to queries to the random oracle that do not amount to guessing any reply correctly. The entropy of verifier messages is typically high in applications,

$p/(\Delta - \epsilon)$，其中 p 是多项式， ϵ 是知识错误。Wikström [12] 将该概念推广到常轮特殊安全协议，并给出了精确且严格的界限。这项工作的具体贡献是针对嫁接协议的一个相应定理。

定理 *1*（非正式）。设（$\mathcal{P}$，$\mathcal{V}$）是一个具有 $2r+1$-消息 $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$-特殊声音协议，对于知识提取器具有声音错误 ϵ_S 和知识错误 ϵ_K，该知识提取器对于任何实例和证明者，以概率 $\Delta > \epsilon_K$使验证者信服，对于常数 c，预期执行协议 $c/(\Delta - \epsilon_K)$ 次。

然后其 $(2\ell + 1)$-消息嫁接协议（$\mathsf{G}[\mathcal{P}], \mathsf{G}[\mathcal{V}]$）对于一个知识提取器存在正确性错误 $\ell\epsilon_S$ 和知识错误 $\ell\epsilon_K$，该知识提取器对于任何实例和证明者，以概率 $\Delta > \ell\epsilon_K$ 说服验证者，预期将执行嫁接协议 $4 \cdot 3^{r+1}\ell \cdot c/(\Delta - \ell\epsilon_K)$ 次。

在应用中，我们可能经常选择协议的参数，以将声音性和知识错误减少 $1/\ell$倍。因此，在实践中，Fiat-Shamir 变换会导致约 $\log \ell + O(1)$ 比特的安全损失。

对符号和概念进行仔细选择，以及一个技术引理，使我们能够有效地将构造提取器的问题简化为在适当定义的格树中找到一个接受基的组合问题，但验证器消息的分布受攻击者影响，不一定均匀。主要的技术挑战是证明这种分布可以被高效地采样。除此之外，来自 [12] 的分析仍然适用，只是细节有所变化。

注记。Attema、Fehr 和 Klooss 最近通知我们，在我们发现之后几个月，他们独立地发现了一个与我们的主要定理相似的定理。有趣的是，他们的工作基于 Attema 等人 [2]，而我们依赖于 Wikström [12] 的工作。因此，在未来的工作中，我们希望理解两种方法的优缺点。

1.2 证明策略

嫁接协议。我们首先基于一个特殊的可靠协议，将匿名交互式Fiat-Shamir证明的计算形式化，将其视为一个嫁接协议的执行。在这样的协议中，验证者允许证明者反复：(1) 生成基本协议的新执行，或 (2) 通过向其嫁接一个额外的轮次来扩展基本协议的现有部分执行。

因此，在执行过程中，任何时刻的转录内容都可以被视为一个部分执行树，每轮增长一条边。我们强调，证明者可以选择每个嫁接轮次的位置，并且这可能取决于树的结构以及到目前为止看到的验证者消息。如果存在一个嵌入的接受转录内容，对应于从根到叶的路径，验证者将接受。

这忠实记录了Fiat-Shamir证明的计算过程，但证明者只能在前一轮执行完毕后才能执行一轮，即它实际上仅限于对随机预言机的查询，而这些查询不涉及正确猜测任何回复。验证者消息的熵在应用中通常很高，

and applying the Fiat-Shamir transform at all to a round with small entropy is pointless. Thus, for such protocols a prover can only guess correctly with exponentially small probability, and the grafting protocol is essentially a faithful model.

Extraction problem for grafting protocols. An extractor for a special sound protocol repeatedly, and recursively: (1) samples an accepting transcript, and (2) samples other accepting transcripts from a prefix of the first. If this fails within reasonable time it gives up and rewinds before restarting the recursive procedure. Additionally, the verifier messages at a given depth are sampled such that the children of each node form a basis of a matroid determined by the protocol.

An extractor that treats the prover as a blackbox must rewind it to extract a suitable tree of transcripts. Rewinding is easy to visualize for interactive protocols, but for a grafting protocol this means that leaves are pruned in the reverse order in which they were grafted to the tree of partial executions. Furthermore, when the protocol is executed with fresh randomness from a partial grafting transcript the tree of partial transcripts may regrow into a differently shaped tree and not only have different verifier messages associated with the nodes.

To understand the added complexity in the extraction problem for grafting protocols it is worthwhile to consider an embedded accepting transcript in a grafting transcript. To rewind the execution of the embedded execution to a given round requires rewinding the execution of the grafting protocol.

The problem is that even if we complete an accepting execution from the rewinded state there is no guarantee that the resulting embedded accepting transcript is a completion of the prefix of the original embedded accepting transcript. Indeed, the prover may spawn a new execution of the special sound protocol, or graft additional rounds to any of the existing partial executions to form a new embedded accepting transcript.

Linearization and grafted sequences. The prover messages are a deterministic function of the verifier messages induced by the prover, which means that we can view: (1) the entire protocol execution, and (2) the verdict of the verifier as a single predicate and focus on the verifier messages.

The added complexity that comes with a tree of partial transcripts is partially superficial, since the the actual execution of the grafting protocol proceeds linearly and the transcript is simply a list of messages that appears in topological order with respect to the tree if we encode the position of a grafted round as an integer index.

Furthermore, although each index for grafting a round is adversarially controlled, it is a deterministic function of the verifier messages thus far in the execution. Thus, the distribution of the list of verifier messages is induced by the prover and can be efficiently sampled. We call the sequence of verifier messages that is generated by this process a *grafted sequence*.

Shadow sequences and sampling. From a *complete* accepting grafted sequence z of the verifier we know the positions of the verifier messages belonging to the corresponding embedded accepting transcript, and the corresponding prover messages can be computed deterministically.

因此，对熵较小的轮次应用Fiat-Shamir变换是毫无意义的。因此，对于此类协议，证明者只能以指数级极小的概率正确猜测，而嫁接协议本质上是一个忠实模型。

嫁接协议的提取问题。一个特殊声音协议的提取器反复地、递归地：(1) 采样一个接受转录本，以及 (2) 从第一个的前缀中采样其他接受转录本。如果在合理的时间内失败，它就会放弃并回滚，然后重新启动递归过程。此外，在给定深度处采样的验证器消息，使得每个节点的子节点形成一个由协议确定的拟阵的基。

一个将证明者视为黑盒的提取器必须回滚它以提取一个合适的转录本树。对于交互式协议，回滚很容易可视化，但对于嫁接协议，这意味着在部分执行树中嫁接的叶子以与它们被嫁接到树中的相反顺序被修剪。此外，当使用来自部分嫁接转录本的最新随机性执行协议时，部分转录本树可能会重新生长成一个不同形状的树，并且不仅与节点相关联的验证器消息不同。

为了理解嫁接协议提取问题中增加的复杂性，考虑一个嵌入在嫁接转录本中的接受转录本是值得的。要将嵌入执行的执行回滚到给定轮次，需要回滚嫁接协议的执行。

问题是，即使我们从回滚状态完成一个接受执行，也无法保证生成的嵌入接受转录文本是原始嵌入接受转录文本前缀的完整实现。实际上，证明者可能会启动特殊安全协议的新执行，或将额外的轮次嫁接到任何现有部分执行上，以形成新的嵌入接受转录文本。

线性化和嫁接序列。证明者的消息是证明者诱导的验证者消息的确定性函数，这意味着我们可以将：(1)整个协议执行，以及(2)验证者的裁决视为单个谓词，并专注于验证者的消息。

部分转录树带来的额外复杂性是表面性的，因为嫁接协议的实际执行是线性的，转录文本仅仅是按树拓扑顺序出现的消息列表，如果我们将嫁接轮次的位置编码为整数索引。

此外，尽管用于生成圆形的每个索引都受到对抗性控制，但它却是执行到目前为止的验证器消息的确定性函数。因此，验证器消息列表的分布由证明者诱导，并且可以被高效地采样。我们称由该过程生成的验证器消息序列为grafted序列。

阴影序列和采样。从一个完整的接受嫁接序列 z 中，我们知道验证器消息的位置属于相应的嵌入接受转录，并且相应的证明者消息可以确定性地计算出来。

Thus, given a grafted sequence z, we can partition it into a *shadow sequence* of the form $(w_1, \ldots, w_r)$, where w_i ends with the ith verifier message of the embedded transcript (except w_r which is slightly different). We call this a shadow sequence, since prefixes are not stable under the addition of elements. More precisely, suppose that z is a grafted sequence with shadow sequence w, and that the shadow prefix $w_{[i]}$ equals the prefix $z_{[k]}$ if we concatenate its components. If z' is a grafted sequence with prefix $z_{[k]}$, then the prefix $w'_{[i]}$ of its shadow sequence may have no common elements with $w_{[i]}$.

We may still think of each shadow element w_i as sampled from a ground set of a *shadow matroid* $\mathbb{M}_i^*$, which inherits the essential combinatorical structure from the corresponding matroid $\mathbb{M}_i$ of the special sound protocol, but the distribution is influenced by the prover.

If we from any prefix $w_{[i]}$ of an accepting shadow sequence w with reasonable probability could sample a complete shadow sequence w', then we would have reduced the extraction problem for grafting protocols to that of basic special sound protocols, i.e., the analysis from [12] would apply with minor syntactical changes.

Sampling shadow sequences. Unfortunately, we can only sample *grafted* sequences directly. Suppose that w is the shadow sequence of grafted sequence z and that the prefix $w_{[i]}$ corresponds to a prefix $z_{[k]}$ of z. Then if a randomly sampled z' conditioned on $z'_{[k]} = z_{[k]}$ is accepting with probability Δ we can certainly sample an accepting grafted sequence z' from the prefix $z_{[k]}$ in time roughly $1/\Delta$, but in general the probability that its shadow sequence w' satisfies $w'_{[i]} = w_{[i]}$ may be arbitrarily low.

Similarly, if we ignore the requirement on acceptance, and focus on keeping the prefix of the shadow sequence it is not hard to see that a random prefix can be extended with conditional probability roughly $1/\ell$ throughout the recursive process.

We show that both properties can be maintained simultaneously throughout an execution with constant probability of failure in each step of the process and thereby allow sampling accepting shaddow sequences that extend the prefixes that appear in the algorithm. Thus, the difference between the extraction problem for a special sound protocol and its grafting protocol is surprisingly small.

2 Background

We need a number of definitions and concepts from [12]. Recall that a matroid $\mathbb{M} = (S, I)$ consists of a ground set S and a set I of subsets of S that is closed downwards and satisfy the independent set exchange property. A basis is a set $B \in I$ such that $B \cup \{x\} \notin I$ for every $x \in S \setminus B$. The rank $\mathsf{rank}(\mathbb{M})$ of a matroid is the unique maximal number of elements in a basis. The span of a set A is defined by $\mathsf{span}(A) = \{x \in S \mid \mathsf{rank}(A \cup \{x\}) = \mathsf{rank}(A)\}$. A flat is a set which is its own span. Section F provides explicit definitions. Throughout d_i denotes the rank of $\mathbb{M}_i$.

A matroid tree $(\{v_0\}, \mathbb{M}_1, \ldots, \mathbb{M}_r)$, where v_0 is an arbitrary singleton, represents the set of verifier messages in a special sound protocol as well as the independence relations needed from a set of accepting transcripts to allow computation of the witness. A subtree is a basis if for each node at depth $i-1$ its children form a basis of $\mathbb{M}_i$.

因此，给定一个嫁接序列 z，我们可以将其划分为一个阴影序列 $(w_1, \ldots, w_r)$ 的形式，其中 w_i 以嵌入转录的 i 个验证器消息结尾（除了 w_r 略有不同）。我们称之为阴影序列，因为前缀在添加元素时不稳定。更精确地说，假设 z 是一个具有阴影序列 w 的嫁接序列，并且如果我们将它的组件连接起来，阴影前缀 $w_{[i]}$ 等于前缀 $z_{[k]}$。如果 z' 是一个具有前缀 $z_{[k]}$ 的嫁接序列，那么它的阴影序列的前缀 $w'_{[i]}$ 可能与 $w_{[i]}$ 没有共同元素。

我们可能仍然认为每个阴影元素 w_i 是从阴影格矩阵 $\mathbb{M}_i^*$ 的基础集合中采样的，该阴影格矩阵继承了特殊声音协议相应格矩阵 $\mathbb{M}_i$ 的基本组合结构，但分布受证明者影响。

如果我们从接受阴影序列 w 的任何前缀 $w_{[i]}$ 以合理的概率能够采样完整的阴影序列 w'，那么我们将把嫁接协议的提取问题简化为基本特殊声音协议的问题，即 [12] 的分析将适用于微小的句法变化。

采样阴影序列。不幸的是，我们只能直接采样嫁接序列。假设 w 是嫁接序列 z 的阴影序列，且前缀 $w_{[i]}$ 对应 z 的前缀 $z_{[k]}$。那么如果条件 $z'_{[k]} = z_{[k]}$ 下随机采样的 z' 以概率 Δ 接受，我们当然可以从前缀 $z_{[k]}$ 在时间大致为 $1/\Delta$ 的时间内采样接受嫁接序列 z'，但通常其阴影序列 w' 满足 $w'_{[i]} = w_{[i]}$ 的概率可能任意低。

类似地，如果我们忽略接受要求，而专注于保持shadow序列的前缀，那么不难看出，一个随机前缀可以通过条件概率大致以 $1/\ell$ 在整个递归过程中扩展。

我们证明，这两种属性可以在整个执行过程中同时保持，每个步骤的失败概率为常数，从而允许采样接受shadow序列，这些序列扩展了算法中出现的前缀。因此，特殊声音协议的提取问题与其嫁接协议之间的差异出人意料地小。

2 背景

我们需要来自 [12]的一些定义和概念。回想一下，一个matroid $\mathbb{M} = (S, I)$由一个基础集 S 和一个向下闭的子集集 I 组成，这些子集满足独立集交换属性。一个基是一个集合 $B \in I$，使得$B \cup \{x\} \notin I$ 对于每个 $x \in S \setminus B$。一个matroid的秩 $\mathrm{rank}(\mathbb{M})$是基中元素的最大唯一数量。一个集合 A 的跨度由$\mathrm{span}(A) = \{x \in S \mid \mathrm{rank}(A \cup \{x\}) = \mathrm{rank}(A)\}$定义。一个平面是一个等于其自身的跨度集合。第F节提供了明确的定义。在整个 d_i 表示 $\mathbb{M}_i$的秩。

一个由 $(\{v_0\}, \mathbb{M}_1, \ldots, \mathbb{M}_r)$ 表示的 matroid 树，其中 v_0 是一个任意单元素，它既代表了特殊安全协议中的验证器消息集合，也代表了从一组接受转录中所需的独立性关系，以便允许计算证物。一个子树是一个基，如果对于深度为 $i-1$ 的每个节点，其子节点形成一个 $\mathbb{M}_i$ 的基。

Definition 1 (Matroid Tree). *The* matroid tree *associated with a list of matroids* $\mathbb{M} = (\{v_0\}, \mathbb{M}_1, \ldots, \mathbb{M}_r)$ *is the vertex-labeled rooted unordered directed tree of depth* r *such that: the root is labeled* v_0 *and every node at depth* $i-1$ *has edges to* $|S_i|$ *children which are uniquely labeled with the elements of the ground set* S_i.

Definition 2 (Basis). *A* basis *of a matroid tree* $\mathbb{M}$ *of depth* r *is a maximal subgraph such that for every* $i \in [r]$ *the set of children of every node at depth* $i-1$ *is a basis of* $\mathbb{M}_i$.

The subdensity captures the fraction of elements of the ground set which is outside a flat. This was introduced in [12] to allow analysis of protocols where verifier messages are chosen from a subset of the algebraic structure which defines the independence sets.

Definition 3 (Subdensity). *Let* $\mathbb{M} = (S, I)$ *be a matroid of rank* d. *Then its* i*th* subdensity *is* $\omega_{\mathbb{M},i}$ *if* $|A|/|S| \leq \omega_{\mathbb{M},i}$ *for every flat* A *of rank* $i-1$, *and it has* maximal subdensity $\omega_{\mathbb{M}} = \omega_{\mathbb{M},d}$.

After abstracting the execution of a protocol and the verdict of the verifier as a predicate ρ on verifier messages the extraction problem amounts to finding a basis of a matroid tree. Let $S = \times_{i \in [r]} S_i$ and $\Delta_\rho(\mathbb{M}) = \Pr[\rho(v) = 1]$, where v is sampled uniformly over a matroid tree $\mathbb{M}$.

Definition 4 (Accepting Basis Extractor). *A probabilistic polynomial time algorithm* $\mathcal{X}_\kappa$ *parametrized by* $\kappa \in \{0,1\}^*$ *is a* $(\epsilon_\kappa, \mathsf{D}_\kappa(\Delta))$-accepting basis extractor *with* extraction error ϵ_κ *for a matroid tree* $\mathbb{M}$, *where* $\mathsf{D}_\kappa(\Delta)$ *for fixed* κ *is a family of distributions on* $\mathbb{N}$ *parametrized by* $\Delta \in [0,1]$, *if for every* $\mathbb{M}$*-predicate* $\rho : S \to \{0,1\}$ *and* $\Delta_\rho(\mathbb{M}) \geq \Delta_0 > \epsilon_\kappa$: $\mathcal{X}_\kappa^{\rho(\cdot)}(\mathbb{M}, \Delta_0)$ *outputs a* ρ*-accepting basis of* $\mathbb{M}$, *where the distribution of the number of* $\rho(\cdot)$*-queries is bounded by* $\mathsf{D}_\kappa(\Delta_0)$.

3 Grafting Protocols

Before we introduce grafting protocols we need some notation. The ith message of the prover is a pair (p_i, a_i), where p_i is the index of a previous verifier message onto which the new branch is grafted and a_i is a prover message of the basic protocol. The verifier always sends its next challenge message immediately, so there is no need for an additional index for the verifier's messages. One round of interaction is therefore always a triple (p_i, a_i, v_{i+1}), where $p_i = 0$ if the prover starts a fresh execution of the basic protocol. Every path in the tree of partial executions of the basic protocol then has the form $(a_{j_1-1}, v_{j_1}, \ldots, a_{j_i-1}, v_{j_i})$, where $p_{j_i} = j_{i-1}$, i.e., it is an embedded transcript of the basic protocol. To ensure that the distribution of verifier messages is correct, the grafting verifier keeps state and samples each message from the appropriate set.

3.1 Functions of Transcripts

After each prover message during an execution of the grafting protocol on common input x the current transcript has the form $\big(x, (p_1, a_1, v_2), \ldots, (p_{i-1}, a_{i-1}, v_i), (p_i, a_i)\big)$ for some i. We call this a *truncated transcript* and denote it by $t_{[i]}$, where t may be a complete or truncated transcript itself. This allows us to define a natural depth function.

定义 *1*（*Matroid* 树）。与一个 *matroid* 列表 $\mathbb{M} = (\{v_0\}, \mathbb{M}_1, .., \mathbb{M}_r)$ 相关的 *matroid* 树是一个深度为 r 的顶点标记的根节点无序有向树，满足：根节点标记为 v_0，并且每个深度为 $i-1$ 的节点都有边连接到 $|S_i|$ 个子节点，这些子节点被唯一标记为基本集合 S_i 的元素。

定义*2*（基础）。一个深度为 r 的 *matroid* 树 $\mathbb{M}$ 的基础是一个极大子图，使得对于每个 $i \in [r]$，深度为 $i-1$ 的每个节点的子集是 $\mathbb{M}_i$ 的基础。

subdensity 捕获了 ground set 中不在平面外的元素的比例。这是在 [12] 中引入的，以允许分析验证器消息从定义独立集的代数结构的子集中选择的协议。

定义*3*（子密度）。设 $\mathbb{M} = (S, I)$ 是一个秩为 d 的 *matroid*。那么它的 i 子密度是 $\omega_{\mathbb{M},i}$ *if* $|A|/|S| \leq \omega_{\mathbb{M}\,i}$ *for*，每个秩为 $i-1$ 的平面 A，并且具有最大子密度 $\omega_{\mathbb{M}} = \omega_{\mathbb{M}\,d}$，

在将协议的执行和验证器的裁决抽象为一个关于验证器消息的谓词 ρ 之后，提取问题归结为找到一个 matroid 树的基础。设 $S = \times_{i \in [r]} S_i$ 和 $\Delta_\rho(\mathbb{M}) = \Pr[\rho(v) = 1]$，其中 v 在 matroid 树 $\mathbb{M}$ 上均匀采样。

定义*4*（接受基础提取器）。一个概率多项式时间算法$\mathcal{X}_\kappa$，以 $\kappa \in \{0,1\}^*$ 为参数，是一个$(\epsilon_\kappa, D_\kappa(\Delta))$-接受基础提取器，具有提取误差 ϵ_κ，用于一个矩阵树 $\mathbb{M}$，其中对于固定的 κ，是一个在 $\mathbb{N}$ 上的分布族，以 $\Delta \in [0,1]$为参数，如果对于每个 $\mathbb{M}$-谓词 $\rho : S \to \{0,1\}$ 和 $\Delta_\rho(\mathbb{M}) \geq \Delta_0 > \epsilon_\kappa$: $\mathcal{X}_\kappa^{\rho(\cdot)}(\mathbb{M}, \Delta_0)$ 输出一个 ρ-接受的 $\mathbb{M}$基础，其中 $\rho(\cdot)$查询的分布被 $\mathsf{D}_\kappa(\Delta_0)$限制。

3 嫁接协议

在我们介绍嫁接协议之前，我们需要一些符号。证明者的第 i条消息是一个对 (p_i, a_i)，其中 p_i 是先前验证者消息的索引，新分支将嫁接到该消息上，而 a_i 是基本协议的证明者消息。验证者总是立即发送其下一个挑战消息，因此验证者的消息不需要额外的索引。因此，每一轮交互总是一个三元组 (p_i, a_i, v_{i+1})，其中 $p_i = 0$ 如果证明者开始基本协议的新执行。基本协议部分执行树中的每条路径的形式是 $(a_{j_1-1}, v_{j_1}, \ldots, a_{j_i-1}, v_{j_i})$，其中 $p_{j_i} = j_{i-1}$ i.e.，，它是一个嵌入的基本协议转录。为了确保验证者消息的分布是正确的，嫁接验证者保持状态，并从适当的集合中采样每条消息。

3.1 文本记录的功能

在grafting协议对公共输入 x 的执行过程中，每次证明者消息后，当前文本记录的形式为 $\big(x\ (p_1, a_1, v_2)\ \ldots\ (p_{i-1}, a_{i-1}, v_i)\ (p_i, a_i)\big)$，，，，对于某些 i。我们称其为截断文本记录，并用 $t_{[i]}$表示，其中 t 本身可以是完整的或截断的文本记录。这使我们能够定义一个自然深度函数。

Definition 5 (Depth Function). *The* depth function δ *takes a truncated transcript of a grafting protocol as input and is defined by*

$$\delta(t_{[i]}) = \begin{cases} 1 & \text{if } p_i = 0 \\ 1 + \delta(t_{[p_i]}) & \text{otherwise} \end{cases} .$$

When the truncated transcript is clear from the context we abuse notation and simply write $\delta(p_i)$ to mean $\delta(t_{[i]})$.

Definition 6 (Index Function). *The* index function $\iota(\cdot)$ *takes a truncated transcript of a grafting protocol as input and is defined by* $\iota(t_{[i]}) = (j_1, \ldots, j_d)$*, where* $d = \delta(t_{[i]})$*,* $j_d = p_i$*, and* $j_l = p_{j_{l+1}}$ *for* $l = d-1, \ldots, 1$*.*

These functions merely gives a way to refer to the unique embedded partial transcript that was most recently extended by a round of interaction. Finally, we introduce notation for extracting the embedded transcript itself using the index function.

Definition 7 (Path Projection). *The* path projection $\tau(\cdot)$ *takes a truncated transcript of a grafting protocol as input and is defined by*

$$\tau(t_{[i]}) = (x, a_{j_1-1}, v_{j_1}, \ldots, a_{j_d-1}, v_{j_d}, a_i) \ , \quad \text{where } (j_1, \ldots, j_d) = \iota(t_{[i]}) \ .$$

Note that if $d = r$, then the embedded transcript is complete and is either accepting or rejecting.

3.2 Grafting Verifier

We give an explicit transformation of a public coin verifier into a grafting verifier for completeness. It is implicit that it rejects if an index p_i provided by the prover is invalid, i.e., if there does not exist a verifier message with the index p_i in the existing partial transcript onto which a round can be grafted.

Without loss of generality we assume that the verifier sends exactly ℓ messages and that the prover's final message corresponds to a final reply of an accepting execution of the basic protocol, if any exists at all.

Definition 8 (Grafting Verifier). *If* $(\mathcal{P}, \mathcal{V})$ *is a* $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$*-special sound protocol, then on common input* x *its* ℓ-grafting verifier $\mathsf{G}[\mathcal{V}]$ *proceeds as follows:*

1. *Initialize an empty table* $T[\cdot, \cdot]$*.*
2. *For* $i = 0, \ldots, \ell - 1$*:*
   (a) *Wait for a message* (p_i, a_i) *from the prover.*
   (b) *If* $T[p_i, a_i] \neq \varnothing$*, then set* $v_{i+1} = T[p_i, a_i]$*, and otherwise choose* $v_{i+1} \in S_{\delta(p_i)}$ *randomly, set* $T[p_i, a_i] = v_{i+1}$*, and hand* v_{i+1} *to the prover.*
3. *Wait for a message* (p_ℓ, a_ℓ) *from the prover.*
4. *Return the verdict* $\mathcal{V}\big(\tau(x, (p_i, a_i, v_{i+1})_{i \in [0, \ell-1]}, p_\ell, a_\ell)\big)$ *of the verifier* $\mathcal{V}$*.*

定义5（深度函数）。深度函数 δ 以*grafting*协议的截断文本记录为输入，其定义为

$$\delta(t_{[i]}) = \begin{cases} 1 & \text{if } p_i = 0 \\ 1 + \delta(t_{[p_i]}) & \text{otherwise} \end{cases} .$$

当截断的转录内容在上下文中清晰时，我们滥用记法，直接写 $\delta(p_i)$来表示 $\delta(t_{[i]})$。

定义 6（索引函数）。索引函数 $\iota(\cdot)$ 以一个嫁接协议的截断转录为输入，并由 $\iota(t_{[i]}) = (j_1, \ldots, j_d)$ 定义，其中 $d = \delta(t_{[i]})$、$j_d = p_i$ 和 $j_l = p_{j_{l+1}}$ *for* $l = d-1 \ \ldots \ 1.$,

这些函数仅提供了一种引用最近由一轮交互扩展的最独特的嵌入部分转录的方法。最后，我们引入了使用索引函数提取嵌入转录本身的记法。

定义 7（路径投影）。路径投影 $\tau(\cdot)$ 以一个嫁接协议的截断转录为输入，并由

$$\tau(t_{[i]}) = (x, a_{j_1-1}, v_{j_1}, \ldots, a_{j_d-1}, v_{j_d}, a_i) \ , \quad \text{where } (j_1, \ldots, j_d) = \iota(t_{[i]}) \ .$$

请注意，如果 $d = r$，则嵌入转录是完整的，要么是接受状态，要么是拒绝状态。

3.2 嫁接验证器

我们给出一个将公开硬币验证器显式转换为接枝验证器的转换，以实现完整性。隐含的是，如果证明者提供的索引 p_i 无效，则它会被拒绝，即如果现有部分会话中不存在一个带有索引 p_i 的验证器消息，无法在其上接枝一轮。

为不失一般性，我们假设验证器发送恰好 ℓ 条消息，并且证明者的最终消息对应于基本协议的接受执行的最终回复（如果存在的话）。

定义 8（接枝验证器）。如果 $(\mathcal{P}, \mathcal{V})$ 是一个 $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$-特殊安全协议，那么在公共输入 x 上，其 ℓ-接枝验证器 $\mathsf{G}[\mathcal{V}]$ 执行如下：

1. 初始化一个空表 $T[\cdot, \cdot]$。
2. 对于 $i = 0 \ \ldots \ \ell - 1$:,
   (a) 等待来自证明者的消息 (p_i, a_i)。
   (b) *If* $T[p_i, a_i] \neq \varnothing$，然后设置 $v_{i+1} = T[p_i, a_i]$，否则选择 $v_{i+1} \in S_{\delta(p_i)}$ 随机地，设置 $T[p_i, a_i] = v_{i+1}$，并将 v_{i+1} 交给证明者。
3. 等待证明者发送消息 (p_ℓ, a_ℓ)。
4. 返回验证者 $\mathcal{V}\big(\tau(x, (p_i, a_i, v_{i+1})_{i \in [0, \ell-1]}, p_\ell, a_\ell)\big)$ 的裁决结果。

Each verifier message is chosen from the ground set associated with the appropriate round in the basic protocol due to the depth function. To avoid grafting more than once at a given index with the *same* prover message the verifier uses a table. This mirrors the same property of a random oracle, i.e., once sampled it returns the same output every time it is queried on the same input.

We are interested in malicious provers which graft branches during the execution, but for completeness we describe in Section D the corresponding honest prover which is merely a wrapper of the honest prover of the basic protocol.

3.3 Grafting Protocols vs Non-interactive Fiat-Shamir Proofs

We may interpret the execution of a grafting protocol as the prover computing a Fiat-Shamir proof in the random oracle model using a random oracle $\mathcal{RO}$ as follows. Relative to the current truncated transcript $t_{[c]}$ a prover message (p_i, a_i) with $i \leq c$ uniquely identifies an embedded transcript $\tau(t_{[i]})$. For this embedded transcript the next verifier message v_{i+1} is independently and uniformly distributed in the appropriate matroid ground set and sampled exactly once.

When the min entropy of the verifier message in each round is high this is essentially equivalent to the computation of a Fiat-Shamir proof, where the next verifier message is defined by $v_{i+1} = \mathcal{RO}(\tau(t_{[i]}))$. Indeed, if the min entropy η is high, then the probability that a prover queries the random oracle in advance at a point partially defined by random verifier messages that it has not yet received is bounded by $\ell 2^{-\eta}$.

In general we cannot expect that it is infeasible to: (1) determine if a prover message is likely to be part of an accepting execution, or (2) use re-randomization to generate arbitrarily many such prover messages from one. This means that a prover can probe up to ℓ verifier messages. Thus, if the min entropy is not exponentially smaller than $1/\ell$ the protocol may loose all soundness, i.e., it is unwise to apply the Fiat-Shamir transform at all.

4 Grafted Sequences

Recall that in [12] the extraction problem is reduced to the problem of extracting an accepting basis of a matroid tree relative a prover predicate that captures both the execution of the protocol and the verdict of the verifier.

We proceed similarly to abstract the extraction of a tree of transcripts of a grafting protocol which correspond to an accepting basis tree in the basic protocol, but in our case the distribution of verifier messages depends on the prover.

A grafting function determines, from the list of verifier messages so far, at which point an additional branch is grafted to a sequence, i.e., given a sequence as input it outputs an integer index of an existing element in the sequence. This abstracts the choice made by the prover in a grafting protocol. A depth function makes explicit the depth at which a branch is grafted.

Definition 9 (Grafting Function). *A function f such that $f(\varnothing) = 0$ and $f(z_1, \ldots, z_i) \in [0, i]$ for every $z_1, \ldots, z_i \in \{0,1\}^*$ and every $i \in \mathbb{N}$ is a* grafting function.

每个验证者消息都根据深度函数从与基本协议中相应轮次关联的基础集合中选择。为了避免在给定索引处使用相同的证明者消息进行一次以上的嫁接，验证者使用一个表。这反映了随机预言机的相同属性，即一旦采样，它每次在相同输入上查询时都会返回相同的输出。

我们对执行期间会添加分支的恶意证明者感兴趣，但为了完整性，我们在第D节中描述了相应的诚实证明者，它仅仅是基本协议中诚实证明者的包装。

3.3 添加分支协议与非交互式Fiat-Shamir证明

我们可以将添加分支协议的执行解释为证明者在随机预言模型中使用随机预言 $\mathcal{RO}$ 计算Fiat-Shamir证明，具体如下。相对于当前截断的会话记录 $t_{[c]}$，证明者消息(p_i, a_i)与 $i \leq c$ 唯一地标识了一个嵌入的会话记录 $\tau(t_{[i]})$。对于这个嵌入的会话记录，下一个验证者消息 v_{i+1} 在适当的格基集合中独立且均匀地分布，并且只采样一次。

当每轮中验证者消息的最小熵较高时，这本质上等同于计算Fiat-Shamir证明，其中下一个验证者消息由 $v_{i+1} = \mathcal{RO}(\tau(t_{[i]}))$定义。实际上，如果最小熵 η 较高，那么证明者在它尚未收到的情况下，在部分由它尚未收到的随机验证者消息定义的点预先查询随机预言的概率被 $\ell 2^{-\eta}$限制。

通常，我们无法期望以下两点不可行：(1) 判断一个证明者消息是否可能是接受执行的一部分，或 (2) 使用重新随机化从一个证明者消息中生成任意多个此类证明者消息。这意味着证明者可以探测高达 ℓ 个验证者消息。因此，如果最小熵不是指数级小于 $1/\ell$，协议可能会完全失去可靠性，即，完全应用 Fiat-Shamir 变换是不明智的。

4 融合序列

回想一下，在 [12] 中，提取问题被简化为提取一个与证明者谓词相对应的 matroid 树的接受基的问题，该证明者谓词捕获了协议的执行和验证者的裁决。

我们类似地抽象出融合协议的文本树提取，这些文本树对应于基本协议中的接受基树，但在我们的情况下，验证者消息的分布取决于证明者。

一个嫁接函数根据目前已有的验证者消息列表，确定在序列的哪个位置嫁接一个额外的分支，即给定一个序列作为输入，它输出序列中一个现有元素的整数索引。这抽象了证明者在嫁接协议中所做的选择。深度函数明确指定了分支嫁接的深度。

定义 *9*（嫁接函数）。一个函数 f，使得 $f(\varnothing) = 0$ 并且 $f(z_1, \ldots, z_i) \in [0, i]$ 对每一个 $z_1 \ldots z_i \in \{0,1\}^*$，以及每一个 $i \in \mathbb{N}$ 是一个嫁接函数。

Definition 10 (Depth Function). *The* depth function δ_f *of a grafting function* f *is defined as follows:*

$$\delta_f(z_1,\ldots,z_i)=\begin{cases}1 & \text{if } f(z_{[i]})=0\\ 1+\delta_f(z_{[f(z_{[i-1]})]}) & \text{otherwise}\end{cases}.$$

A grafted sequence is an abstraction of a transcript of a grafted protocol where the verifier messages are explicit, and the prover messages are implicit.

Definition 11 (Grafted Sequence). *An* $(\mathbb{M},f)$-grafted sequence of length ℓ, *where* $\mathbb{M}=(\mathbb{M}_1,\ldots,\mathbb{M}_r)$ *is matroid tree with* $\mathbb{M}_i=(S_i,I_i)$ *is a sequence* $z=(z_1,\ldots,z_\ell)$ *such that* $\delta_f(z_{[i-1]})\le r$ *and* $z_i\in S_{\delta_f(z_{[i-1]})}$ *for every* $i\in[\ell]$. *We denote the set of* $(\mathbb{M},f)$*-grafted sequences of length* ℓ *by* $G_{\mathbb{M},f,\ell}$.

Similarly to how we extracted indices from a grafted protocol transcript we extract indices of the verifier messages of an (implicitly defined) embedded truncated transcript of the basic protocol. It is not meaningful to define a path projection since the prover messages are defined by the complete grafted sequence.

Definition 12 (Index Function). *The* index function ι_f *takes a grafted sequence* $z\in G_{\mathbb{M},f,\ell}$ *as input and outputs indices* $(j_1,\ldots,j_d)$ *defined by* $j'=f(z)$, $d=\delta_f(z_{[j']})$, $j_d=j'$, *and* $j_i=f(z_{[j_{i+1}-1]})$ *for* $i=d-1,\ldots,1$.

4.1 Shadow Sequences

We introduce shadow sequences and shadow matroids as a conceptual step to emphasize the similarity with the analysis in [12].

Definition 13 (Shadow Sequence). *If* z *is a grafted sequence over* $G_{\mathbb{M},f,\ell}$, $(j_1,\ldots,j_d)=\iota_f(z)$, $j_0=0$, *and* $w_i=(z_{j_{i-1}+1},\ldots,z_{j_i})$ *for* $i\in[1,d-1]$, *and* $w_d=(z_{j_{d-1}+1},\ldots,z_\ell)$, *then* $\sigma_f(z)=(w_1,\ldots,w_d)$ *is its* shadow sequence.

The last shadow element ends at index ℓ and not index j_d, so there may be some additional elements beyond the last embedded verifier message of the basic protocol. This is necessary, but it does not impact the analysis technically.

For every grafted sequence z there is a shadow sequence $w=\sigma_f(z)$ and we may view a predicate ρ over grafted sequences as a predicate ρ^* over shadow sequences. The last component of the ith shadow element is an element from S_i. Thus, the ith shadow element is contained in the following matroid.

Definition 14 (Shadow Matroid). *If* $\mathbb{M}=(S,I)$ *is a matroid, then its* shadow matroid $\mathbb{M}^*=(S^*,I^*)$ *is defined by* $S^*=\{0,1\}^*\times S\times\{0,1\}^*$ *and letting* C *be an independence set in* I^* *if and only if* $B=\{b\mid(a,b,c)\in C\}$ *is an independence set in* I *and* $|C|=|B|$.

Intuitively, we would like to think of a shadow sequence as a redundant representation of a list of verifier messages in the basic special sound protocol, but the prefixes/postfixes influence the implicitly defined prover messages and the output of the grafting function, so this remains an intuitive view.

定义 *10*（深度函数）。一个嫁接函数 f 的深度函数 δ_f 定义如下：

$$\delta_f(z_1,\ldots,z_i)=\begin{cases}1 & \text{if } f(z_{[i]})=0\\ 1+\delta_f(z_{[f(z_{[i-1]})]}) & \text{otherwise}\end{cases}.$$

一个嫁接序列是对一个嫁接协议的转录本的抽象，其中验证器消息是显式的，而证明者消息是隐式的。

定义*11*（嫁接序列）。一个长度为 ℓ的$(\mathbb{M},f)$-嫁接序列，其中$\mathbb{M}=(\mathbb{M}_1,\ldots,\mathbb{M}_r)$是格树，$\mathbb{M}_i=(S_i,I_i)$是一个序列 $z=(z_1,\ldots,z_\ell)$，使得对于每个 $i\in[\ell]$，有 $\delta_f(z_{[i-1]})\le r$和 $z_i\in S_{\delta_f(z_{[i-1]})}$。我们用$G_{\mathbb{M}、f、\ell}$表示长度为 ℓ 的$(\mathbb{M},f)$-嫁接序列的集合。

类似于我们从嫁接协议转录本中提取索引的方式，我们提取（隐式定义的）基本协议嵌入截断转录本中验证器消息的索引。由于证明者消息由完整的嫁接序列定义，因此定义路径投影没有意义。

定义 *12*（索引函数）。索引函数 ι_f 接收一个嫁接序列 $z\in G_{\mathbb{M},f,\ell}$ 作为输入，并输出由 $j'=f(z)$ $d=\delta_f(z_{[j']})$, ,$j_d=j'$, 和 $j_i=f(z_{[j_{i+1}-1]})$ *for* $i=d-1\ \ldots\ 1$, , 定义的下标 $(j_1,\ldots,j_d)$。

4.1 阴影序列

我们引入阴影序列和阴影格，作为强调与 [12] 中分析相似性的概念性步骤。

定义 *13*（阴影序列）。如果 z 是一个在 $G_{\mathbb{M}\ f\ \ell}$ $(j_1,\ldots,j_d)=$, , ,$\iota_f(z)$, $j_0=0$, 和 $w_i=(z_{j_{i-1}+1},\ldots,z_{j_i})$ *for* $i\in[1,d-1]$ 上的嫁接序列，并且 $w_d=(z_{j_{d-1}+1},\ldots,z_\ell)$, 则 $\sigma_f(z)=(w_1,\ldots,w_d)$ 是它的阴影序列。

最后一个影子元素在索引 ℓ 处结束，而不是在索引 j_d 处，因此可能存在一些额外的元素，超出了基本协议中最后一个嵌入的验证器消息。这是必要的，但技术上不影响分析。

对于每个嵌入序列 z，都有一个影子序列 $w=\sigma_f(z)$，我们可以将关于嵌入序列的谓词 ρ 视为关于影子序列的谓词 ρ^*。第 i 个影子元素的最后一个组件是一个来自 S_i 的元素。因此，第 i 个影子元素包含在以下矩阵中。

定义 *14*（影子矩阵）。如果 *If* $\mathbb{M}=(S,I)$ 是一个矩阵，那么它的影子矩阵 $\mathbb{M}^*=(S^*,I^*)$ 是通过 $S^*=\{0,1\}^*\times S\times\{0,1\}^*$ 定义的，并让 C 是 I^* 中的一个独立集，当且仅当 *f* $B=\{b\mid(a,b,c)\in C\}$ 是 I 中的一个独立集并且$|C|=|B|$。

直观上，我们希望将影子序列视为基本特殊声音协议中验证者消息列表的一种冗余表示，但前缀/后缀会影响隐式定义的证明者消息以及嫁接函数的输出，因此这仍然是一种直观的看法。

4.2 Grafting Function and Predicate of a Prover

We define a grafting function in terms of a grafting protocol and use the definition of a prover predicate from [12], which is restated below with adapted notation for easy reference.

Definition 15 (Grafting Function of Prover). *The* grafting function $f[\mathcal{P}^*, \mathcal{V}]$ of $\mathcal{P}^*$ *for an ℓ-grafting protocol of a public-coin protocol $(\mathcal{P}, \mathcal{V})$ and common input x is defined as follows: On input $z = (z_1, \ldots, z_i)$, simulate $(\mathcal{P}^*, \mathsf{G}[\mathcal{V}])$ using z as the random tape for $\mathsf{G}[\mathcal{V}]$ until $\mathcal{P}^*$ outputs its ith message (p_i, a_i) and output p_i.*

Definition 16 (Prover Predicate). *The* prover $\mathbb{M}$-predicate $\rho[\mathcal{P}^*, \mathcal{V}, x]$ of $\mathcal{P}^*$ *for the ℓ-grafting protocol of a public-coin protocol $(\mathcal{P}, \mathcal{V})$ and common input x is defined by $\rho[\mathcal{P}^*, \mathcal{V}, x](z) = \langle \mathcal{P}^*, \mathsf{G}[\mathcal{V}]_z \rangle(x)$, where $z = (z_1, \ldots, z_\ell)$.*

4.3 Random Grafted Sequences

Suppose that $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ is a matroid tree and let f be a grafting function. A random variable over $G_{\mathbb{M},f,\ell}$ representing the verifier messages of an execution of a grafting protocol is readily defined by stipulating that each verifier message is uniformly and independently distributed over a ground set identified by the depth function.

Definition 17 (Random Grafted Sequence). *The distribution of a random grafted sequence Z over $G_{\mathbb{M},f,\ell}$ is defined by $\mathsf{P}_{Z_i|Z_{[i-1]}}\left(\cdot \,|z_{[i-1]}\right) = |S_{\delta_f(z_{[i-1]})}|^{-1}$.*

Although each element Z_i is uniformly and independently distributed, the sequence is not necessarily uniformly distributed, since the choice of ground set is determined by previous elements and the grafting function. Consequently, the distribution of the shadow sequence $W = \sigma_f(Z)$ is not necessarily uniform.

5 Random Shadow Sequences

In each recursive call of the extractor the probability that the current prefix of a shadow sequence leads to an accepting sequence is assumed to be some quantity Δ, but we must also be able to efficiently sample extensions of the prefix.

The problem is that even if the prefix has probability Δ to lead to an accepting grafted sequence it may be the case that the resulting grafted sequence does not have the same prefix as a shadow sequence. Indeed, partitioning of the grafted sequence into a shadow sequence is determined by the grafted sequence as a whole. Conversely, if we focus on sampling a shadow sequence with a given prefix, then the acceptance probability under this conditioning may be significantly lower than Δ. Thus, we must prove that a random prefix has both properties at once with reasonable probability. We formalize the property we need below.

Definition 18 (Extendable Shadow Prefix). *Let $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ be a matroid tree, let f be a grafting function, let Z be a random grafted sequence over $G_{\mathbb{M},f,\ell}$,*

4.2 插接函数和证明者的谓词

我们根据嫁接协议定义一个嫁接函数，并使用来自 [12]，的证明者谓词的定义，该定义使用调整后的符号重新陈述如下，以便于参考。

定义 *15*（证明者的插接函数）。证明者 $f[\mathcal{P}^*, \mathcal{V}]$ 的 $\mathcal{P}^*$ 对于一个公共币协议 *(P, V)* 的 ℓ-插接协议和公共输入 x 的插接函数 $f[\mathcal{P}^*, \mathcal{V}]$ 定义如下：在输入 $z = (z_1, \ldots, z_i)$ 时，使用 z 作为 *G*[$\mathcal{V}$] 的随机带模拟 $(\mathcal{P}^*, \mathsf{G}[\mathcal{V}])$，直到 $\mathcal{P}^*$ 输出其 i 次消息 *(p_i, a_i)* 并输出 p_i。

定义 *16*（证明者谓词）。证明者 $\mathbb{M}$ 的 $\rho[\mathcal{P}^*, \mathcal{V}, x]$ 对于公共币协议 *(P, V)* 的 ℓ-插接协议和公共输入 x 的谓词 $\mathbb{M}$ 定义为 $\rho[\mathcal{P}^*, \mathcal{V}, x](z) = \langle \mathcal{P}^*, \mathsf{G}[\mathcal{V}]_z \rangle(x)$，其中 $z = (z_1, \ldots, z_\ell)$。

4.3 随机嫁接序列

假设 $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ 是一个拟阵树，并令 f 是一个嫁接函数。一个在 $G_{\mathbb{M}\ f\ \ell}$ 上的随机变量，表示一个嫁接协议执行过程中的验证者消息，可以通过规定每个验证者消息在由深度函数确定的基集上均匀且独立地分布而轻易定义。

定义 *17*（随机嫁接序列）。随机嫁接序列 Z 在 $G_{\mathbb{M}}$、$_f$、$_\ell$ 上的分布由 $P_{Z_i|Z_{[i-1]}}\left(\cdot \,|z_{[i-1]}\right) = |S_{\delta_f(z_{[i-1]})}|^{-1}$ 定义。

尽管每个元素 Z_i 是均匀且独立地分布的，但序列不一定是均匀分布的，因为基集的选择由先前元素和嫁接函数决定。因此，影子序列 $W = \sigma_f(Z)$ 的分布不一定是均匀的。

5个随机阴影序列

在提取器的每次递归调用中，阴影序列当前前缀引导至接受序列的概率被假定为某个量 Δ，但我们还必须能够高效地采样前缀的扩展。

问题是，即使前缀有概率 Δ 引导至接受嫁接序列，也可能导致最终嫁接序列不具有与阴影序列相同的相同前缀。实际上，将嫁接序列划分为阴影序列的划分由整个嫁接序列决定。相反，如果我们专注于采样具有给定前缀的阴影序列，那么在此条件下的接受概率可能显著低于 Δ。因此，我们必须证明随机前缀同时具有这两种属性的概率是合理的。我们将在下面形式化所需的属性。

定义 *18*（可扩展阴影前缀）。设 $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ 是一个拟阵树，设 f 是一个嫁接函数，设 Z 是一个在 $G_{\mathbb{M},f,\ell}$, 上的随机嫁接序列

define $J = \iota_f(Z)$, and $W = \sigma_f(Z)$, and let $\rho : G_{\mathbb{M},f,\ell} \to \{0,1\}$ be a predicate. Define for every $i \in [0, r-1]$, $w \in [W_{[i]}]$, and $\beta \in (0,1)$:

$$\zeta_w^\rho = \Pr\left[\rho(Z) = 1 \,\middle|\, Z_{[k]} = w\right] \tag{1}$$

$$\theta_w^\rho = \Pr\left[J_i = k \,\middle|\, \rho(Z) = 1 \wedge Z_{[k]} = w\right] \tag{2}$$

$$\xi_\rho(w, \Delta, \beta) = \left(\zeta_w^\rho \geq \Delta \wedge \theta_w^\rho \geq \beta^2/\ell\right) \ . \tag{3}$$

We can always sample a complete grafting sequence Z starting with $w_{[i]}$ and if $\xi_\rho(w_{[i]}, \Delta, \beta) = 1$, then we have $\rho(Z) = 1 \wedge J_i = k$ with probability at least $\beta^2\Delta/\ell$. Below we show that this implies $\rho^*(W) = 1$ and $W_{[i]} = w_{[i]}$. Thus, we need roughly $\ell/(\beta^2\Delta)$ sampled grafted sequences starting from $w_{[i]}$ to find an accepting shadow sequence starting from $w_{[i]}$. We maintain a sufficient acceptance probability by application of the following well known lemma, which is proven in Section E.

Lemma 1 (Markov Conditioning). *If $H = (X, Y)$ is a random variable, E is an event in $[H]$, $\delta_x = \Pr_H[E \,|\, X = x]$, and $\Pr_H[E] \geq \Delta$, then $\Pr_H[\delta_X < \alpha\Delta \,|\, E] \leq \alpha$.*

5.1 Coinciding Indices

It should be clear that if z and z' share a prefix $z_{[k]}$ corresponding to a prefix $w_{[i]}$ of the shadow sequence w of z, and the ith element of the shadow sequence w' of z' end at index k, then $w'_{[i]} = w_{[i]}$.

Lemma 2 (Pinching). *For every $z, z' \in G_{\mathbb{M},f,\ell}$ and every $i \in [1, r-1]$, with $j = \iota_f(z)$ and $w = \sigma_f(z)$, and similarly for j' and w', we have*

$$z'_{[j_i]} = z_{[j_i]} \ \textit{and} \ j'_i = j_i \quad \Longrightarrow \quad j'_{[i]} = j_{[i]} \ \textit{and} \ w'_{[i]} = w_{[i]} \ .$$

Proof. If we define $p_t = f(z_{[t-1]})$ for $t \in [\ell]$ and similarly for p'_t and $z'_{[t-1]}$, then by assumption $p'_{[i]} = p_{[i]}$. Thus, if $j'_i = j_i$, then $j'_{[i]} = j_{[i]}$ which implies that $w'_{[i]} = w_{[i]}$.

Suppose that we sample a grafted sequence z and let $w_{[i]}$ be a prefix of its shadow sequence w, which viewed as a prefix of the grafted sequence has the form $z_{[k]}$ for some k. If we sample a fresh completion $z'_{[k+1,\ell]}$ of $z_{[k]}$, and define $w' = \sigma_f(z_{[k]}, z'_{[k+1,\ell]})$ and $j' = \iota_f(z_{[k]}, z'_{[k+1,\ell]})$, then Lemma 2 says that it is sufficient to require that $j'_i = j_i$ to guarantee that $w'_{[i]} = w_{[i]}$. The next lemma is used to prove that over the random choice of $w_{[i]}$ this happens with reasonable probability.

Lemma 3 (Coinciding Indices). *Let $Z = (Z_1, \ldots, Z_\ell)$ be a random variable, let $\iota : [Z] \to [0, \ell-1]$ be a function, define $K = \iota(Z)$, $X = (Z_1, \ldots, Z_K)$, and let Y be independently distributed with $\mathsf{P}_{Y|X}(\cdot|x) = \mathsf{P}_{Z_{k+1},\ldots,Z_\ell|X}(\cdot|x)$, where x has length k. If we define $\theta_x = \Pr[\iota(X,Y) = k \,|\, X = x]$, then for every $\beta \in (0, 1/2)$:*

$$\Pr\left[\theta_X < \beta^2/\ell\right] \leq 2\beta \ . \tag{4}$$

定义 $J = \iota_f(Z)$，定义 $W = \sigma_f(Z)$，并让 $\rho : G_{\mathbb{M}\ f\ \ell} \to \{0,1\}$，，成为一个谓词。对于每一个 $i \in [0, r-1]$、$w \in [W_{[i]}]$ 和 $\beta \in$ *(0, 1)*：

$$\zeta_w^\rho = \Pr\left[\rho(Z) = 1 \,\middle|\, Z_{[k]} = w\right] \tag{1}$$

$$\theta_w^\rho = \Pr\left[J_i = k \,\middle|\, \rho(Z) = 1 \wedge Z_{[k]} = w\right] \tag{2}$$

$$\xi_\rho(w, \Delta, \beta) = \left(\zeta_w^\rho \geq \Delta \wedge \theta_w^\rho \geq \beta^2/\ell\right) \ . \tag{3}$$

我们可以始终采样一个完整的嫁接序列 Z，以 $w_{[i]}$ 开头，如果 $\xi_\rho(w_{[i]}, \Delta, \beta) = 1$，那么我们就有 $\rho(Z) = 1 \wedge J_i = k$，概率至少为 $\beta^2\Delta/\ell$。下面我们证明这蕴含 $\rho^*(W) = 1$ 和 $W_{[i]} = w_{[i]}$。因此，我们需要大约 $\ell/(\beta^2\Delta)$ 个从 $w_{[i]}$ 开始采样的嫁接序列，以找到一个从 $w_{[i]}$ 开始的接受阴影序列。我们通过应用以下著名引理来保持足够的接受概率，该引理在 E 节中证明。

引理 *1*（马尔可夫条件）。*If* $H = (X, Y)$ 是一个随机变量，E 是 $[H]$ $\delta_x = \Pr_H[E \,|\, X = x]$ 中的一个事件，如果 $\Pr_H[E] \geq \Delta$，那么 $Pr_H[\delta_X < \alpha\Delta \,|\, E] \leq \alpha$。

5.1 重合索引

应该很清楚，如果 z 和 z' 共享一个前缀 $z_{[k]}$，该前缀对应于影子序列 w 的前缀 $w_{[i]}$，并且影子序列 w' 的第 i 个元素在索引 k 处结束，那么 $w'_{[i]} = w_{[i]}$。

引理 *2*（捏合）。对于任意 z、$z' \in G_{\mathbb{M}\ f\ \ell}$、$i \in [1, r-1]$，以及满足 $j = \iota_f(z)$ 和 $w = \sigma_f(z)$ 的 j' 和 w'，我们有

$$z'_{[j_i]} = z_{[j_i]} \ \textit{and} \ j'_i = j_i \quad \Longrightarrow \quad j'_{[i]} = j_{[i]} \ \textit{and} \ w'_{[i]} = w_{[i]} \ .$$

证明。如果我们定义 $p_t = f(z_{[t-1]})$ 对于 $t \in [\ell]$，并且类似地定义 p'_t 对于 $z'_{[t-1]}$，那么根据假设 $p'_{[i]} = p_{[i]}$。因此，如果 $j'_i = j_i$，那么 $j'_{[i]} = j_{[i]}$，这意味着 $w'_{[i]} = w_{[i]}$.

假设我们采样一个嫁接序列 z，并让 $w_{[i]}$ 是其影子序列 w 的一个前缀，将其视为嫁接序列的前缀时，其形式为 $z_{[k]}$，对于某个 k。如果我们采样一个全新的完成 $z'_{[k+1,\ell]}$ of $z_{[k]}$，并定义 $w' = \sigma_f(z_{[k]}, z'_{[k+1,\ell]})$ 和 $j' = \iota_f(z_{[k]}, z'_{[k+1,\ell]})$，那么引理 2 表明，只需要求 $j'_i = j_i$ 即可保证 $w'_{[i]} = w_{[i]}$。下一个引理用于证明，在随机选择 $w_{[i]}$ 时，这种情况以合理的概率发生。

引理 *3*（重合索引）。设 $Z = (Z_1, \ldots, Z_\ell)$ 是一个随机变量，设 $\iota : [Z] \to [0, \ell-1]$ 是一个函数，定义 $K = \iota(Z)$ $X = (Z_1, \ldots, Z_K)$，，并且设 Y 独立地分布与 $\mathsf{P}_{Y|X}(\cdot|x) = \mathsf{P}_{Z_{k+1} \ldots Z_\ell|X}(\cdot|x)$，，其中 x 的长度为 k。如果我们定义 $\theta_x = \Pr[\iota(X,Y) = k \,|\, X = x]$，那么对于每一个 $\beta \in (0, 1/2)$:

$$\Pr\left[\theta_X < \beta^2/\ell\right] \leq 2\beta \ . \tag{4}$$

Proof. By definition we have

$$\mathsf{P}_{X,K}(x,k) = \mathsf{P}_{Z_{[k]}|K}(x|k)\,\mathsf{P}_K(k) \quad , \tag{5}$$

where it is understood that $\mathsf{P}_{X,K}(x,k) = 0$ if the length of x is not equal to k. Thus, to sample x we may: sample a length k, sample $z_{[k]}$ as a prefix of a complete sequence z conditioned on $\iota(z) = k$, and set $x = z_{[k]}$. Furthermore, from independence we have

$$\Pr[\iota(X,Y) = k \,|X = x] = \Pr\left[\iota(Z) = k \,\middle|Z_{[k]} = x\right] \tag{6}$$

which means that $\theta_x = \Pr\left[\iota(Z) = k \,\middle|Z_{[k]} = x\right]$.

If we let $\beta \in (0,1)$ and define $B = \{k \mid \mathsf{P}_K(k) < \beta/\ell\}$, then we trivially have $\Pr[K \in B] < \sum_{k\in[0,\ell-1]} \beta/\ell = \beta$. For every $k \notin B$ we have $\Pr[\iota(Z) = k] = \Pr[K = k] \geq \frac{\beta}{\ell}$ from the definitions of K and the set B. For $k \notin B$ and every $\alpha \in (0,1)$ Lemma 1 then implies that

$$\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \leq \alpha \quad , \tag{7}$$

which implies that

$$\Pr[\theta_X < \alpha\beta/\ell] = \sum_{k\in[0,\ell-1]} \mathsf{P}_K(k)\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \tag{8}$$

$$\leq \Pr[K \in B] + \sum_{k\notin B} \mathsf{P}_K(k)\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \tag{9}$$

$$\leq \beta + \alpha\sum_{k\notin B} \mathsf{P}_K(k) \leq \alpha + \beta \quad . \tag{10}$$

The proof is completed by setting $\alpha = \beta$.

5.2 Extendable Shadow Sequence

The following theorem follows from the two lemmas above and the union bound.

Theorem 2 (Extendable Shadow Sequence). *Let $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ be a matroid tree, let f be a grafting function, let Z be a random grafted transcript over $G_{\mathbb{M},f,\ell}$, and define $J = \iota_f(Z)$ and $W = \sigma_f(Z)$. Let $\rho : G_{\mathbb{M},f,\ell} \to \{0,1\}$ be a predicate. For every $i \in [r-1]$, $w \in [W_{[i-1]}]$ such that $\zeta_w^\rho \geq \Delta$, $\alpha \in (0,1)$, and $\beta \in (0,(1-\alpha)/2)$.*

$$\Pr\left[\xi_\rho(W_{[i]}, \alpha\Delta, \beta) = 1 \,\middle|\rho^*(W) = 1, W_{[i-1]} = w\right] \geq 1 - \alpha - 2\beta \quad . \tag{11}$$

Proof. We define the random variable (J_i, X), by

$$\mathsf{P}_{J_i,X}(\cdot) = \mathsf{P}_{J_i,W_{[i]}|W_{[i-1]}}(\cdot\,|w) \tag{12}$$

In other words, X effectively captures the distribution of the ith shadow element and its ending index conditioned on the $i-1$ previous shadow elements in w. Next we define $H = (X,Y)$ by defining an independently distributed random variable Y

$$\mathsf{P}_{Y|X}(\cdot\,|x) = \mathsf{P}_{Z_{[k+1,\ell]}|Z_{[k]}}(\cdot\,|x) \quad . \tag{13}$$

证明。根据定义我们有

$$\mathsf{P}_{X,K}(x,k) = \mathsf{P}_{Z_{[k]}|K}(x|k)\,\mathsf{P}_K(k) \quad , \tag{5}$$

在理解$\mathsf{P}_{X\ K}(x,k) = 0$ 的情况下，如果 x 的长度不等于 k。因此，要采样 x，我们可以：采样长度 k，采样$z_{[k]}$ 作为完整序列 z的前缀，条件为 $\iota(z) = k$，并设置$x = z_{[k]}$。此外，从独立性我们得到

$$\Pr[\iota(X,Y) = k \,|X = x] = \Pr\left[\iota(Z) = k \,\middle|Z_{[k]} = x\right] \tag{6}$$

这意味着 $\theta_x = \Pr\left[\iota(Z) = k \,\middle|Z_{[k]} = x\right]$。

如果我们让 $\beta \in (0,1)$ 并定义 $B = \{k \mid \mathsf{P}_K(k) < \beta/\ell\}$，那么我们显然有 $\Pr[K \in B] < \sum_{k\in[0,\ell-1]} \beta/\ell = \beta$。对于每个 $k \notin B$，根据 K 和集合 B的定义，我们有 $\Pr[\iota(Z) = k] = \Pr[K = k] \geq \frac{\beta}{\ell}$。对于 $k \notin B$ 和每个$\alpha \in (0,1)$，引理1意味着

$$\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \leq \alpha \quad , \tag{7}$$

这意味着

$$\Pr[\theta_X < \alpha\beta/\ell] = \sum_{k\in[0,\ell-1]} \mathsf{P}_K(k)\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \tag{8}$$

$$\leq \Pr[K \in B] + \sum_{k\notin B} \mathsf{P}_K(k)\Pr\left[\theta_{Z_{[k]}} < \alpha\beta/\ell \,\middle|\iota(Z) = k\right] \tag{9}$$

$$\leq \beta + \alpha\sum_{k\notin B} \mathsf{P}_K(k) \leq \alpha + \beta \quad . \tag{10}$$

通过设置 $\alpha = \beta$来完成证明。

5.2 可扩展阴影序列

以下定理源于上述两个引理和并集界限。

定理2（可扩展阴影序列）。设 $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ 是一个拟阵树，f 是一个嫁接函数，Z 是在$G_{\mathbb{M}\ f\ \ell}$上的随机嫁接转录，...，并定义 $J = \iota_f(Z)$ 和 $W = \sigma_f(Z)$。设 $\rho : G_{\mathbb{M}\ f\ \ell} \to \{0,1\}$，...是一个谓词。对于每一个$i \in [r-1]$ $w \in [W_{[i-1]}]$，满足 $\zeta_w^\rho \geq \Delta$ $\alpha \in (0,1)$，...，并且 $\beta \in (0,(1-\alpha)/2)$。

$$\Pr\left[\xi_\rho(W_{[i]}, \alpha\Delta, \beta) = 1 \,\middle|\rho^*(W) = 1, W_{[i-1]} = w\right] \geq 1 - \alpha - 2\beta \quad . \tag{11}$$

证明。我们通过以下方式定义随机变量 (J_i, X),

$$\mathsf{P}_{J_i,X}(\cdot) = \mathsf{P}_{J_i,W_{[i]}|W_{[i-1]}}(\cdot\,|w) \tag{12}$$

换句话说，X 有效地捕获了 i个阴影元素的分布及其结束索引，条件是 $i-1$ 之前的阴影元素在 w中。接下来我们通过定义一个独立分布的随机变量 Y来定义$H = (X,Y)$

$$\mathsf{P}_{Y|X}(\cdot\,|x) = \mathsf{P}_{Z_{[k+1,\ell]}|Z_{[k]}}(\cdot\,|x) \quad . \tag{13}$$

The random variable Y represents the sampling of a completion of a grafting sequence starting with x. Finally, we define $K = \#(X)$, i.e., K is the index of what we expect to be the last element of the ith element of the shadow sequence.

By assumption $\Pr[\rho(H) = 1] \geq \Delta$. Thus, if we define $\zeta_x^\rho = \Pr[\rho(H) = 1 \,|\, X = x]$, then from Lemma 1 we have the bound $\Pr[\zeta_X^\rho < \alpha\Delta \,|\, \rho(H) = 1] \leq \alpha$. If we define $\theta_x^\rho = \Pr[J_i = k \,|\, \rho(H) = 1 \wedge X = x]$, where k is the length of x, then Lemma 3 implies that $\Pr\left[\theta_X^\rho < \beta^2/\ell \,\middle|\, \rho(H) = 1\right] \leq 2\beta$. The union bound finally gives

$$\Pr[\theta_X^\rho \geq \alpha\Delta \wedge \zeta_X^\rho \geq \beta \,|\, \rho(H) = 1] \geq 1 - \alpha - 2\beta \ , \tag{14}$$

which concludes the proof.

6 Accepting Basis Extractor for Shadow Sequences

To construct an extractor for grafting protocols we first show that shadow sequences can be sampled. This trivially gives a basic extractor and a basic sampler of shadow sequences corresponding to the basic algorithms in [12]. The recursive extractor follows by syntactic changes, since it is is defined in terms of the expected value and tail bound for each recursive call and the tail bounds do not change.

6.1 Shadow Sampler

Theorem 2 says that from a suitable prefix $w_{[i]}$ of shadow sequences we can sample a complete accepting shadow sequence w that keeps the prefix intact, but we also need to ensure that $w_{i+1} \in \mathbb{M}_{i+1}^* \backslash \mathsf{span}(B^*)$, where B^* is the shadow version of an independent set $B \in I_{i+1}$, to ensure that we end up with a basis of $\mathbb{M}^*$. This can be accomplished by sampling every grafting element at depth $i+1$ from $I_{i+1} \setminus \mathsf{span}(B)$ instead of from I_{i+1}. There are at most ℓ such elements in a sequence so the statistical distance between this modified distribution and the original is at most $\ell\omega_{\mathbb{M}_{i+1}}$. Theorem 2 then implies the following lemma, which is proven in Section B for completeness.

Lemma 4 (Shadow Sampler). *There exists a sampler $\mathcal{S}_{\alpha_1}^{f,\rho}$ of shadow sequences such that for every grafting function f, predicate ρ, and $\alpha_1 \in (0,1)$, and any input $(\mathbb{M}, w_{[i]}, B, \Delta_0)$ such that $\xi_\rho\left(w_{[i]}, \Delta_0, \beta_0\right) = 1$ with $\Delta_0 > \ell\omega_{\mathbb{M}_{i+1}}$:*

1. *the distribution of the number of calls to ρ is bounded by $\mathsf{Geo}(\beta_0^2\Delta_1'/\ell)$, and*
2. *the output w has prefix $w_{[i]}$, $\rho^*(w) = 1$, and $w_{i+1} \in \mathbb{M}_{i+1} \setminus \mathsf{span}(B^*)$, and $\Pr\left[\xi_\rho\left(w_{[i+1]}, \Delta_1, \beta_1\right) = 1\right] \geq \beta_1$,*

where $\Delta_1' = \Delta_0 - \ell\omega_{\mathbb{M}_{i+1}}$, $\Delta_1 = \alpha\Delta_1'$, and $\beta_1 = \frac{1}{3}(1-\alpha_1)$:

We need to change the syntax slightly to accomodate for prefixes needed to sample correctly, but the shadow sampler makes it trivial to construct a basic sampler $\mathcal{S}_\alpha^{f,\rho}$ that from an extendable prefix samples the next shadow element conditioned on acceptance. We similarly denote by $\mathcal{B}_\alpha^{f,\rho}$ the basic extractor for shadow sequences that takes an input $w_{[r-1]}$ and invokes the shadow sampler d_r times with the parameter α, storing the new elements in an initially empty set B^*, to find accepting a set of transcripts with the prefix $w_{[r-1]}$ such that their rth elements form a basis over $\mathbb{M}_r^*$.

随机变量 Y 表示从 x开始的接枝序列完成的采样。最后，我们定义 $K = \#(X)$，即 K 是我们期望的阴影序列中 i个元素中最后一个元素的索引。

根据假设 $\Pr[\rho(H) = 1] \geq \Delta$。因此，如果我们定义 $\zeta_x^\rho = \Pr[\rho(H) = 1 \,|\, X = x]$，那么根据引理 1，我们得到界限 $\Pr[\zeta_X^\rho < \alpha\Delta \,|\, \rho(H) = 1] \leq \alpha$。如果我们定义 $\theta_x^\rho = \Pr[J_i = k \,|\, \rho(H) = 1 \wedge X = x]$，其中 k 是 x 的长度，那么引理 3 意味着 Pr $\left[\theta_X^\rho < \beta^2/\ell \,\middle|\, \rho(H) = 1\right] \leq 2\beta$。最后，并集界限给出

$$\Pr[\theta_X^\rho \geq \alpha\Delta \wedge \zeta_X^\rho \geq \beta \,|\, \rho(H) = 1] \geq 1 - \alpha - 2\beta \ , \tag{14}$$

这就完成了证明。

6 接受基础提取器用于影子序列

为了构建用于嫁接协议的提取器，我们首先证明影子序列可以被采样。这直接给出了一个基本的提取器和一个基本的影子序列采样器，对应于 [12] 中基本算法。递归提取器通过语法变化得到，因为它是以每个递归调用的期望值和尾部界限来定义的，而尾部界限没有改变。

6.1 影子采样器

定理2指出，从一个合适的影子序列前缀 $w_{[i]}$ 中，我们可以采样一个完整的接受影子序列 w，该序列会保留前缀不变，但我们还需要确保 $w_{i+1} \in \mathbb{M}_{i+1}^* \backslash \mathsf{span}(B^*)$，其中 B^* 是独立集 $B \in I_{i+1}$ 的影子版本，以确保最终得到 $\mathbb{M}^*$ 的基。这可以通过从 $I_{i+1} \setminus \mathrm{span}(B)$ 而不是从 I_{i+1} 中采样深度 $i+1$ 的每个嫁接元素来实现。序列中最多有 ℓ 个这样的元素，因此这种修改后的分布与原始分布之间的统计距离最多为 $\ell\omega_{\mathbb{M}_{i+1}}$。定理2进而引出以下引理，该引理在B节中为完整性而证明。

引理*4*（影子采样器）。存在一个影子序列采样器 $\mathcal{S}_{\alpha_1}^{f,\rho}$，对于每个嫁接函数 f、谓词 ρ 和 $\alpha_1 \in (0,1)$，以及任何满足 $\xi_\rho\left(w_{[i]}, \Delta_0, \beta_0\right) = 1$ 且有 $\Delta_0 > \ell\omega_{\mathbb{M}_{i+1}}$ 的输入 $(\mathbb{M}, w_{[i]}, B, \Delta_0)$:

1. ρ 的调用次数分布被 $\mathsf{Geo}(\beta_0^2\Delta_1'/\ell)$限制，并且
2. 输出 w 有前缀 $w_{[i]}$、$\rho^*(w) = 1$，以及 $w_{i+1} \in \mathbb{M}_{i+1} \backslash span(B^*)$，并且 $\Pr\left[\xi_\rho\left(w_{[i+1]}, \Delta_1, \beta_1\right) = 1\right] \geq \beta_1$,

where $\Delta_1' = \Delta_0 - \ell\omega_{\mathbb{M}_{i+1}}$, $\Delta_1 = \alpha\Delta_1'$, and $\beta_1 = \frac{1}{3}(1-\alpha_1)$:

我们需要稍微调整语法以适应采样所需的前缀，但阴影采样器使得构建基本采样器 $\mathcal{S}_\alpha^{f,\rho}$ 变得非常简单，该采样器从可扩展的前缀中采样下一个阴影元素，条件是接受。我们同样用 $\mathcal{B}_\alpha^{f,\rho}$ 表示用于阴影序列的基本提取器，它接受输入 $w_{[r-1]}$ 并使用参数 α 调用阴影采样器 d_r 次，将新元素存储在一个初始为空的集合 B^* 中，以找到具有前缀 $w_{[r-1]}$ 的接受转录集，使得它们的 r 元素在 $\mathbb{M}_r^*$ 上形成基。

6.2 Accepting Basis Extractor

The recursive extractor $\mathcal{R}_\kappa[\mathcal{R}]$, parametrized by a parameter κ, and making recursive calls to $\mathcal{R}$, is virtually identical to that in [12], since it is defined in terms of expected values and tail bounds of a recursive call or the basic extractor. Analytically the situation is equivalent to the original analysis of the basic strategy except for three changes: (1) $\ell\omega_{\mathbb{M}_i}$ replaces the subdensity $\omega_{\mathbb{M}_i}$, (2) we loose a factor 3 for each recursive call due to the threefold use of the union bound, and (3) the expected running time of the basic extractor increases by a factor of $\ell/(1-\alpha_{r-1})^2$. Thus, if we set $\nu_i = 1/\alpha_i$, then we may simply restate the main theorem from [12] with these changes, but we provide a proof in Section C.

We consider families of distributions $\mathsf{D}(s,\Delta)$ parametrized by $s \in \mathbb{N}^+$ and $\Delta \in [0,1]$, which satisfy a tail bound of the form $\Pr\left[X \geq k\mu_{\mathsf{D}(s,\Delta)}\right] \leq t_s^{\mathsf{D}}(k)$, where X is distributed according to $\mathsf{D}(s,\Delta)$ and $\mu_{\mathsf{D}(s,\Delta)}$ is the expected value of $\mathsf{D}(s,\Delta)$. For compound geometric distributions we have $t_s^{\mathsf{CG}}(k) = e^{-(k-1-\ln k)s}$ from [12].

Theorem 3 (Extractor). *For every $\nu_1,\ldots,\nu_{r-1} \in (1,\infty)$ with $\nu_i \geq \nu_{i+1}$ there exist parameters $\kappa_i = (\alpha_i,\lambda_i)$ such that the algorithm $\mathcal{X}_\kappa = \mathcal{R}_{\kappa_1}[\mathcal{R}_{\kappa_2}[\cdots\mathcal{R}_{\kappa_{r-2}}[\mathcal{B}]\cdots]]$, is a $(\epsilon_0, \mathsf{D}_\kappa(\Delta_0))$-accepting basis extractor for shadow matroid tree of $\mathbb{M}$ where:*

$$\epsilon_0 = \ell \sum_{i\in[r]} \omega_{\mathbb{M}_i} \prod_{j\in[i-1]} \nu_j \qquad \textit{(extraction error)} \tag{15}$$

$$\mu_{\mathsf{D}_0(\Delta_0)} \leq \ell \cdot \frac{c_0 \prod_{j\in[r]} d_j}{\Delta_0 - \epsilon_0} \qquad \textit{(expected number of queries)} \tag{16}$$

$$t_{d_1}^{\mathsf{D}_0}(k) \leq t_{d_1}^{\mathsf{CG}}(k) \quad \textit{for } k > 1\ , \qquad \textit{(tail bound)} \tag{17}$$

where the constant c_0 is defined by

$$c_0 = 3^{r+1} \frac{\nu_{r-1}^2}{(\nu_{r-1}-1)^2} \prod_{i\in[r-1]} \frac{\nu_i^2}{(\nu_i - 1)} \cdot \min_{k_i\in(0,1)} \left\{ \frac{k_i}{h_{d_i}^{\mathsf{CG}}(k_i)} \right\} \ . \tag{18}$$

7 Interpretation

We refer the reader to [12] for an in depth intuitive interpretation of the overall recursive formulas. To see that the extraction error ϵ_0 is tight up to a factor r first recall that it upper bounds the soundness error. Then consider a protocol such that: (1) guessing the verifier message of the special sound protocol in any round is necessary to convince the verifier of a false statement, and (2) that the prover can determine if a guess is correct before the execution is continued. For such a protocol the soundness error is roughly $\epsilon = \sum_{i=1}^r \omega_{\mathbb{M}_i}$. In the grafting protocol the prover may probe any round independently with ℓ queries, so the soundness error is bounded by $\ell \max_{i\in[r]}\{\omega_{\mathbb{M}_i}\} \geq \epsilon_0/r$.

When $\Delta_0 \gg \epsilon_0$ the factor ℓ in the extraction error can be ignored and if we set $\nu_i = 2$ the running time is $4 \cdot 3^{r+1}\ell$ times the running time in the interactive case. Consider a prover that effectively executes the basic special sound protocol without

6.2 接受基础提取器

递归提取器 $\mathcal{R}_\kappa[\mathcal{R}]$，由参数 κ 参数化，并递归调用 $\mathcal{R}$，与 [12]，中的几乎相同，因为它是在递归调用或基本提取器的期望值和尾部界限的术语中定义的。从分析的角度来看，情况与基本策略的原始分析等效，除了三个变化：(1) $\ell\omega_{\mathbb{M}_i}$ 替换了子密度 $\omega_{\mathbb{M}_i}$，(2) 由于三次使用并集界限，每次递归调用会损失一个因数 3，(3) 基本提取器的预期运行时间增加了 $\ell/(1-\alpha_{r-1})^2$ 倍。因此，如果我们设置 $\nu_i = 1/\alpha_i$，则可以简单地用这些变化重新陈述来自 [12] 的主要定理，但我们已在 C 节中提供了证明。

我们考虑一族分布 $\mathsf{D}(s,\Delta)$，该族分布由 $s \in \mathbb{N}^+$ 和 $\Delta \in [0,1]$，参数化，并满足形式为 $\Pr\left[X \geq k\mu_{\mathsf{D}(s,\Delta)}\right] \leq t_s^{\mathsf{D}}(k)$ 的尾部界限，其中 X 服从 $\mathsf{D}(s,\Delta)$ 分布，$\mu_{\mathsf{D}(s,\Delta)}$ 是 $\mathsf{D}(s,\Delta)$ 的期望值。对于复合几何分布，我们有 $t_s^{\mathsf{CG}}(k) = e^{-(k-1-\ln k)s}$ 从 [12]。

定理 3（提取器）。对于每个 $\nu_1 \ldots \nu_{r-1} \in (1,\infty)$，，当 $\nu_i \geq \nu_{i+1}$ 时，存在参数 $\kappa_i = (\alpha_i,\lambda_i)$，使得算法 $\mathcal{X}_\kappa = \mathcal{R}_{\kappa_1}[\mathcal{R}_{\kappa_2}[\cdots\mathcal{R}_{\kappa_{r-2}}[\mathcal{B}]\cdots]]$ 是一个 $(\epsilon_0, \mathsf{D}_\kappa(\Delta_0))$-接受基础提取器，用于 $\mathbb{M}$ 的影子拟阵树，其中：

$$\epsilon_0 = \ell \sum_{i\in[r]} \omega_{\mathbb{M}_i} \prod_{j\in[i-1]} \nu_j \qquad \textit{(提取错误)} \tag{15}$$

$$\mu_{\mathsf{D}_0(\Delta_0)} \leq \ell \cdot \frac{c_0 \prod_{j\in[r]} d_j}{\Delta_0 - \epsilon_0} \qquad \textit{(预期查询数量)} \tag{16}$$

$$t_{d_1}^{\mathsf{D}_0}(k) \leq t_{d_1}^{\mathsf{CG}}(k) \quad \textit{for } k > 1\ , \qquad \textit{(尾部界限)} \tag{17}$$

其中常数 c_0 由定义

$$c_0 = 3^{r+1} \frac{\nu_{r-1}^2}{(\nu_{r-1}-1)^2} \prod_{i\in[r-1]} \frac{\nu_i^2}{(\nu_i - 1)} \cdot \min_{k_i\in(0,1)} \left\{ \frac{k_i}{h_{d_i}^{\mathsf{CG}}(k_i)} \right\} \ . \tag{18}$$

7 解释

我们向读者介绍 [12]，以进行深入直观地解释整体递归公式。要看到提取误差 ϵ_0 紧密到因子 r，首先回想它上界了可靠性误差。然后考虑一个协议，使得：(1) 在任何轮次中猜测特殊可靠性协议的验证者消息是让验证者相信一个错误陈述所必需的，以及 (2) 证明者可以在执行继续之前确定一个猜测是否正确。对于这种协议，可靠性误差大约是 $\epsilon = \sum_{i=1}^r \omega_{\mathbb{M}_i}$。在嫁接协议中，证明者可以独立地对任何轮次进行探测，使用 ℓ 查询，因此可靠性误差被 $\ell \max_{i\in[r]}\{\omega_{\mathbb{M}_i}\} \geq \epsilon_0/r$ 所限制。

当 $\Delta_0 \gg \epsilon_0$ 提取误差中的 ℓ 因子可以忽略，并且如果我们将 $\nu_i = 2$ 设置为交互情况下运行时间的 $4 \cdot 3^{r+1}\ell$ 倍时。考虑一个证明者，它在没有接枝任何分叉的情况下，直到第（$\Delta_0 \gg \epsilon_0$）轮才探测 ℓ 轮 $\nu_i = 2$ 消息，然后在一个随机选择的分叉执行中精确完成最后一轮。那么在 $4 \cdot 3^{r+1}\ell$ 轮中进行分叉需要大约 个样本。因此， 因子是必要的，并且运行时间相对紧凑，直到一个 因子。

grafting any forks until the $(r-2)$th round where it probes $\ell-r-3$ round $r-1$ messages before completing the last round in exactly one randomly chosen partial execution. Then forking in round $r-2$ requires roughly ℓ samples. Thus, the factor ℓ is necessary and the running time is relatively tight up to a factor $4 \cdot 3^{r+1}$.

To summarize, our results show that for any constant r and any $(2r+1)$-message special sound protocol the application of the Fiat-Shamir heuristic comes at a cost of a factor ℓ in extraction error and $O(\ell)$ extraction time, respectively, but with the same type of distribution as in the interactive case. In practice the matroid subdensities can typically be decreased by a factor of $1/\ell$ by a different choice of parameters for the protocol and cancel the effect on the extraction error at modest cost in efficiency. Thus, the loss of security from the application of the Fiat-Shamir heuristic in practice is typically no more than $\log \ell + O(1)$ bits of security which is intuitively appealing.

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grafting 任何分叉，直到第 $(r-2)$ 轮，它探测 $\ell-r-3$ 轮 $r-1$ 消息，然后在一个随机选择的分叉执行中精确完成最后一轮。那么在 $r-2$ 轮中进行分叉需要大约 ℓ 个样本。因此，ℓ 因子是必要的，并且运行时间相对紧凑，直到一个 $4 \cdot 3^{r+1}$ 因子。

总而言之，我们的结果表明，对于任何常数 r 和任何 $(2r+1)$-消息特殊声音协议，应用 Fiat-Shamir 启发式方法在提取误差上会付出 ℓ 的代价，在提取时间上会付出 $O(\ell)$ 的代价，但分布类型与交互式情况相同。在实践中，通过选择不同的协议参数，矩阵子密度通常可以降低 $1/\ell$ 的比例，并在效率上付出适度成本来消除对提取误差的影响。因此，在实际应用中，Fiat-Shamir 启发式方法带来的安全损失通常不超过 $\log \ell + O(1)$ 比特，这在直观上很有吸引力。

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A Definitions

We recall the definitions introduced in [12].

Definition 19 (Accepting Basis). *A basis B of a matroid tree $\mathbb{M}$ is ρ-*accepting *for an $\mathbb{M}$-predicate ρ if $\rho(v) = 1$ for each path v of maximal length in B.*

Definition 20 (Accepting Transcript Tree). *A rooted unordered directed tree T with vertex labels $\ell(\cdot)$ is an* accepting transcript tree *for $\mathcal{V}$ if every leaf has depth r and for every path $(u_0, \ldots, u_r)$ in T: $(v_{u_0}, a_{u_0}, \ldots, v_{u_r}, a_{u_r})$ is accepting, where $\ell(u_i) = (v_{u_i}, a_{u_i})$.*

Definition 21 (Challenge Tree). *The* challenge tree $\mathbb{V}(T)$ *of an accepting transcript tree T with vertex labels $\ell(\cdot)$ has the same nodes and vertices, but labels defined by $\ell'(u) = v$, where $\ell(u) = (v, a)$.*

Definition 22 (Special Soundness). *A $(2r+1)$-message public coin-protocol $(\mathcal{P}, \mathcal{V})$ is $((\mathbb{M}_1, \ldots, \mathbb{M}_r), p)$-*special-sound *for an NP relation R, where $\mathbb{M}_i = (S_i, I_i)$ is a matroid, if the ith message of $\mathcal{V}$ is chosen randomly from S_i, and there exists a* witness extraction algorithm *$\mathcal{W}$ that given an accepting transcript tree T such that $\mathbb{V}(T)$ is basis subtree of $(\{x\}, \mathbb{M}_1, \ldots, \mathbb{M}_r)$ outputs a witness w such that $(x, w) \in \mathsf{R}$ in time p.*

B Proof of Lemma 4

The extractor in [12] repeatedly samples complete lists of accepting verifier messages with a slight bias to guarantee independence properties. We intend to essentially execute the original extractor with a shadow predicate ρ^* over a shadow matroid tree $\mathbb{M}^*$.

The original analysis assumes that the lists of verifier messages are sampled uniformly from a matroid, but what is actually necessary for the analysis to work is that they are sampled identically as in the protocol. Thus, nothing prevent us from invoking a modified extractor for our shadow matroid tree, provided that the verifier messages are sampled with the right distribution. Consider the following algorithm.

Definition 23 (Grafted Sequence Sampler). *The* grafted sequence sampler Z^f *takes as input a tuple $(\mathbb{M}, \ell, z, b, B)$, where f is a grafting function, $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ is a matroid tree with $\mathbb{M}_i = (S_i, I_i)$, $z \in G_{\mathbb{M},f,k}$, and $B \in I_b$ is not a basis and proceeds as follows.*

1. *For $i = k+1, \ldots, \ell$:*

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A 定义

我们回顾一下在 [12] 中引入的定义。

定义 *19*（接受基础）。一个 *matroid* 树 B 的基 $\mathbb{M}$ 对于每个在 B 中的最大长度路径 v，是 ρ-接受的基础，对于 $\mathbb{M}$-谓词 ρ *if* $\rho(v) = 1$。

定义 *20*（接受转录树）。一个带顶点标签 $\ell(\cdot)$ 的有根无序有向树 T 是 $\mathcal{V}$ 的接受转录树，如果每个叶子深度为 r，并且对于 T 中的每条路径 $(u_0, \ldots, u_r)$：$(v_{u_0}, a_{u_0}, \ldots, v_{u_r}, a_{u_r})$ 是接受的，其中 $\ell(u_i) = (v_{u_i}, a_{u_i})$。

定义 *21*（挑战树）。接受转录树 T (T) 的挑战树 $\mathbb{V}(\ell)$（顶点标签 $(\ell'(u) = v)$）具有相同的节点和顶点，但标签由 $\ell(u) = (v, a)$ 定义，其中 $\ell(u) = (v, a)$。

定义*22*（特殊正确性）。一个 $(2r+1)$-消息公共硬币协议$(\mathcal{P}, \mathcal{V})$对于*NP*关系*R*是 $((\mathbb{M}_1, \ldots, \mathbb{M}_r)\ p)$，-特殊的，如果 $\mathbb{M}_i = (S_i, I_i)$ 是一个拟阵，当 i个消息从 $\mathcal{V}$ 中随机选择，并且存在一个证言提取算法 $\mathcal{W}$，给定一个接受转录树 T，使得 $\mathbb{V}(T)$是 $(\{x\}, \mathbb{M}_1, \ldots, \mathbb{M}_r)$ 的基树，输出证言 w，使得 $(x, w) \in \mathsf{R}$ 在时间 p内完成。

B 引理4的证明

[12] 中的提取器反复采样接受验证器消息的完整列表，带有轻微的偏差以保证独立性属性。我们打算本质上用阴影谓词 ρ^* 在阴影拟阵树 $\mathbb{M}^*$上执行原始提取器。

原始分析假设验证器消息的列表是从拟阵中均匀采样的，但实际上分析要起作用所必需的是它们与协议中一样地被采样。因此，只要验证器消息被以正确的分布采样，就没有什么能阻止我们从阴影拟阵树中调用修改后的提取器。考虑以下算法。

定义*23*（嫁接序列采样器）。嫁接序列采样器Z^f 以元组$(\mathbb{M}, \ell, z, b, B)$为输入，其中 f是一个嫁接函数，$\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ 是一个具有 $\mathbb{M}_i = (S_i, I_i)$ $z \in G_{\mathbb{M}\ f\ k}$的拟阵树，$B \in I_b$ 不是基，并按以下方式执行。

1. 对于 $i = k+1\ \ldots\ \ell$:，，

(a) *Compute* $d = \delta_f(z)$ *and sample* z_i *randomly, in* $S_d \setminus \mathsf{span}(B)$ *if* $d = b$*, and in* S_d *otherwise.*
(b) *Append* z_i *to* z.

2. *Return* z.

The running time of the algorithm is, apart from sampling in the complement of $\mathsf{span}(B)$, identical to executing the protocol, i.e., its running time corresponds to evaluating the predicate ρ if we ignore the cost of sampling verifier messages.

The algorithm is used below to sample an accepting shadow sequence which has a prefix $w_{[i-1]}$ and we need w_i to not be contained in a set $B^* \in I_b^*$, but at the time of sampling the grafted elements we do not know which sample from S_i will determine independence in I_b^*. Thus, we make sure that all grafted elements from S_b that make up w_i are from $S_b \setminus B$ instead, where B is the projection of B^* to their middle elements. It may seem that this approach introduces an unnecessarily large error in the distribution of the output, i.e., roughly $\ell\omega_{\mathbb{M}_b}$ instead of $\omega_{\mathbb{M}_b}$, but this seems unavoidable. Next we use the grafted sequence sampler to implement a shadow sampler.

Definition 24 (Shadow Sampler). *The* shadow sampler *algorithm* $\mathsf{W}^{f,\rho}$*, where* f *is a grafting function and* ρ *is a predicate, takes as input a tuple* $(\mathbb{M}, \ell, w_{[i]}, B)$*, where* $\mathbb{M} = (\mathbb{M}_0, \ldots, \mathbb{M}_r)$ *is a matroid tree,* $\rho : G_{\mathbb{M},f,\ell} \to \{0,1\}$*,* $w_{[i]}$ *is a prefix of a shadow sequence corresponding to a grafted sequence* $z_{[k]} \in G_{\mathbb{M},f,k}$*, and* $B \in I_{i+1}$ *is not a basis. Repeat:*

1. *Compute* $z = \mathsf{Z}^f(\mathbb{M}, \ell, z_{[k]}, i+1, B)$ *and set* $j = \iota_f(z)$ *and* $w' = \sigma_f(z)$.
2. *If* $\rho(z) = 1$ *and* $j_i = k$*, then return* w'.

Let z be the grafted sequence sampled by $\mathsf{W}^{f,\rho}$ such that $w' = \sigma_f(z)$ is returned, and set $j = \iota_f(z)$. By construction $w_{[i]}$ is a prefix of w' viewed as grafted sequences and it only returns if $j_i = k$. Thus, Lemma 2 implies that $w'_{[i]} = w_{[i]}$. Furthermore, $\mathsf{W}^{f,\rho}$ only returns if $\rho(z) = 1$ which implies that $\rho^*(w') = 1$. Finally, every element from S_{i+1} is sampled from the subset $S_{i+1} \setminus B$, which implies that $w_{i+1} \in S_{i+1}^* \setminus B^*$. This proves the first claim.

If $\xi_\rho(w_{[i]}, \Delta_0, \beta_0) = 1$, then in each iteration the probability of returning is at least $\beta_0^2\Delta_0/\ell$. Thus, the distribution of the number of calls to ρ is bounded according to the second claim. The third claim follows directly from Theorem 2.

C Proof of Theorem 3

We now have the subroutines needed to derive a recursive extractor from the construction in [12] almost by syntactic changes. The only essential difference is that all algorithms need the complete prefix of a partial shadow sequence as input to sample completions with the right distribution.

We need to modify the notion of an accepting basis extractor to allow for the additional parameter ℓ and the parameter β from Theorem 2.

(a) 计算 $d = \delta_f(z)$ 并随机采样 z_i，如果 $d = b$，则在 $S_d \setminus$ *span(B)* 中，否则 S_d 在
(b) 将 z_i 添加到 z。

2. 返回 z。

该算法的运行时间，除了在 $\mathrm{span}(B)$ 的补集中进行采样外，与执行协议相同，即其运行时间对应于评估谓词 ρ，如果我们忽略采样验证器消息的成本。

算法如下用于采样一个接受阴影序列，该序列以 $w_{[i-1]}$ 为前缀，我们需要 w_i 不包含在集合 $B^* \in I_b^*$ 中，但在采样嫁接元素时，我们不知道 S_i 中的哪个样本将决定 I_b^* 中的独立性。因此，我们确保所有构成 w_i 的来自 S_b 的嫁接元素都来自 $S_b \setminus B$，其中 B 是 B^* 到它们中间元素的投影。这似乎会引入一个不必要的较大误差，即输出分布中约为 $\ell\omega_{\mathbb{M}_b}$ 而不是 $\omega_{\mathbb{M}_b}$，但这似乎不可避免。接下来我们使用嫁接序列采样器来实现阴影采样器。

定义 *24*（阴影采样器）。阴影采样器算法 $W^{f,\rho}$，其中 f 是一个嫁接函数，ρ 是一个谓词，它以元组 $(\mathbb{M}, \ell, w_{[i]}, B)$ 作为输入，其中 $\mathbb{M} = (\mathbb{M}_0, \ldots, \mathbb{M}_r)$ 是一个拟阵树，$\rho : G_{\mathbb{M}\ f\ \ell} \to \{0,1\}$，，，$w_{[i]}$ 是对应于嫁接序列 $z_{[k]} \in G_{\mathbb{M}\ f\ k}$，，，的阴影序列的前缀，并且 $B \in I_{i+1}$ 不是一个基。重复：

1. 计算 $z = Z^f(\mathbb{M}, \ell, z_{[k]}, i+1, B)$ 并设置 $j = \iota_f(z)$ 和 $w' = \sigma_f(z)$.
2. 如果 $\rho(z) = 1$ 且 $j_i = k$，则返回 w'.

设 z 为由 $\mathsf{W}^{f\ \rho}$ 采样的嫁接序列，使得返回 $w' = \sigma_f(z)$，并设置 $j = \iota_f(z)$。根据构造，$w_{[i]}$ 是视为嫁接序列的 w' 的前缀，并且只有在 $j_i = k$ 时才返回。因此，引理 2 意味着 $w'_{[i]} = w_{[i]}$。此外，$\mathsf{W}^{f\ \rho}$，只有在 $\rho(z) = 1$ 时才返回，这意味着 $\rho^*(w') = 1$。最后，S_{i+1} 中的每个元素都从子集 $S_{i+1} \setminus B$ 中采样，这意味着 $w_{i+1} \in S_{i+1}^* \setminus B^*$。这证明了第一个断言。

我f $\xi_\rho(w_{[i]}, \Delta_0, \beta_0) = 1$，然后在每次迭代中返回的概率至少为$\beta_0^2\Delta_0/\ell$。因此，调用 ρ 次数的分布根据第二个断言是有界的。第三个断言直接来自定理2。

C 定理3的证明

现在我们有了从 [12] 中的构造中通过语法变化推导出递归提取器的子程序。唯一的关键区别是所有算法都需要部分阴影序列的完整前缀作为输入，以采样具有正确分布的完成项。

我们需要修改接受基提取器的概念，以允许额外的参数 ℓ 以及来自定理 2 的参数 β。

Definition 25 (Accepting Basis Extractor). *A probabilistic polynomial time algorithm $\mathcal{X}_\kappa$ parametrized by $\kappa \in \{0,1\}^*$ is a $(\epsilon_\kappa, \mathsf{D}^\ell_\kappa(\Delta,\beta))$-*accepting basis extractor *with* extraction error ϵ_κ *for the matroid tree $\mathbb{M}' = (\{w_{[i]}\}, \mathbb{M}^*_{i+1}, \ldots, \mathbb{M}^*_r)$, where $\mathsf{D}^\ell_\kappa(\Delta,\beta)$ for fixed κ is a family of distributions on $\mathbb{N}$ parametrized by $\ell \in \mathbb{N}$, $\Delta \in [0,1]$, $\beta \in (0,1)$, if for every grafting function f and predicate $\rho : G_{\mathbb{M},f,\ell} \to \{0,1\}$ the following holds.*

If $\Delta_i > \epsilon_\kappa$, $\beta_i > 0$, and $\xi_\rho(w_{[i]}, \Delta_i, \beta_i) = 1$, then $\mathcal{X}^{f,\rho}_\kappa(\mathbb{M}, \ell, w_{[i]}, \Delta_i, \beta_i)$ outputs a ρ-accepting basis of $\mathbb{M}'$, where the distribution of the number of $\rho(\cdot)$-queries is bounded by $\mathsf{D}^\ell_\kappa(\Delta_i, \beta_i)$.

Definition 26 (Recursive Extractor). *Let $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ be a matroid tree and assume that $\mathcal{R}$ is a $(\epsilon_i, \mathsf{D}^\ell_i(\Delta,\beta))$-accepting basis extractor for matroid trees of the form $(\{w_{[i]}\}, \mathbb{M}^*_{i+1}, \ldots, \mathbb{M}^*_r)$. The* recursive extractor *$\mathcal{R}_\kappa[\mathcal{R}]$, where $\kappa = (\alpha_i, \lambda_i)$ and $\alpha_i, \lambda_i \in (0,1)$ proceeds as follows on input $(\mathbb{M}, \ell, w_{[i-1]}, \Delta_{i-1}, \beta_{i-1})$.*

1. *Set $\Delta_i = \alpha_i(\Delta_{i-1} - \ell\omega_{\mathbb{M}_i})$, $\beta_i = \frac{1}{3}(1-\alpha_i)$, $k = k^{\mathsf{D}_i}(\lambda_i)$, and $\mu = \mu_{\mathsf{D}^\ell_i(\Delta_i,\beta_i)}$.*
2. *Set $B^* = \varnothing$ and $T = \varnothing$.*
3. *While $|B^*| < d_i$:*
   (a) *Compute $w = \mathcal{S}^{f,\rho}_{\alpha_i}(\mathbb{M}, \ell, w_{[i-1]}, B^*, \Delta_{i-1}, \beta_{i-1})$.*
   (b) *Extract subtree $t = \mathcal{R}^{f,\rho}(\mathbb{M}, \ell, w_{[i]}, \Delta_i, \beta_i)$, but interrupt the execution and set $t = \perp$ if it attempts to make more than $k\mu$ queries.*
   (c) *If $t \neq \perp$, then set $B^* = B^* \cup \{w_i\}$ and $T = T \cup \{t\}$.*
4. *Return the accepting basis tree T.*

The following lemma and corollary follows mutatatis mutandi from the corresponding proof in [12], where we use indices to illustrate the similarity with the original recursive formulas.

Lemma 5 (Recursive Extractor). *The algorithm $\mathcal{R}_\kappa[\mathcal{R}]$ is a $(\epsilon_{i-1}, \mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1}))$-accepting basis extractor, where $\epsilon_{i-1} = \epsilon_i/\alpha_i + \ell\omega_{\mathbb{M}_i}$ and*

$$\mathcal{G}_{\mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1})}(z) = \prod_{i=1}^{d_i} \mathcal{G}_{\mathsf{Geo}(\beta_i\lambda_i)}\big(\mathcal{G}_{\mathsf{Geo}(\beta^2_{i-1}\Delta_{i-1}/\ell)}(z) z^{k^{\mathsf{D}_i}(\lambda_i)\mu_{\mathsf{D}^\ell_i(\Delta_i,\alpha_i)}}\big) \ , \quad (19)$$

defined by $\beta_i = \frac{1}{3}(1-\alpha_i)$ and $\Delta_i = \alpha_i(\Delta_{i-1} - \ell\omega_{\mathbb{M}_i})$.

Corollary 1 (Recursive Extractor). *The distribution $\mathsf{D}^\ell_{i-1}(\Delta,\beta)$ satisfies*

$$\mu_{\mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1})} = \frac{3d_i}{(1-\alpha_i)\lambda_i}\left(\frac{\ell}{\beta^2_{i-1}\Delta_i} + k^{\mathsf{D}_i}(\lambda_i)\mu_{\mathsf{D}^\ell_i(\Delta_i,\alpha_i)}\right) \quad (20)$$

$$t^{\mathsf{D}_{i-1}}_{d_i}(k) \le t^{\mathsf{CG}}_{d_i}(k) \quad \textit{for } k \in (1,\infty) \ . \quad (21)$$

If $\beta^2_{i-1} \ge \beta^2_i\alpha_i$, then the term $\ell/(\beta^2_{i-1}\Delta_i)$ can be dropped by observing that the initial sample can be reused in the recursive call and recursive calls are slightly more expensive. When $\beta_i = \frac{1}{3}(1-\alpha_i)$ this is always the case, since $(1-\alpha_{i-1})^2 \ge \alpha_i(1-\alpha_i)^2$ for $\alpha_{i-1}, \alpha_i \in (0,1)$.

Thus, the only change in the expected value compared to the basic case in [12] is a factor of three in each recursive call (and there are $r-1$ levels of recursion), and that the expected value for the basic extractor is increased by a factor of $3^2\ell/(1-\alpha_{r-1})^2$. Setting $\nu_i = 1/\alpha_i$ gives the theorem.

定义 *25*（接受基提取器）。一个由 $\kappa \in \{0,1\}^*$ 参数化的概率多项式时间算法 $\mathcal{X}_\kappa$ 是一个 *$(\epsilon_\kappa, D^\ell_\kappa(\Delta,\beta))$*-接受基提取器，用于 *matroid* 树 $\mathbb{M}' = (\{w_{[i]}\}, \mathbb{M}^*_{i+1}, \ldots, \mathbb{M}^*_r)$，其中对于固定的 κ，$\mathsf{D}^\ell_\kappa(\Delta,\beta)$ 是一个在 $\mathbb{N}$ 上的分布族，由 $\ell \in \mathbb{N}$、$\Delta \in [0,1]$、$\beta \in (0,1)$ 参数化，如果对于每一个嫁接函数 f 和谓词 $\rho : G_{\mathbb{M}\,f\,\ell} \to \{0,1\}$，满足以下条件。

如果 $\Delta_i > \epsilon_\kappa$、$\beta_i > 0$ 和 $\xi_\rho(w_{[i]}$、Δ_i、$\beta_i) = 1$，那么 $\mathcal{X}^{f\ \rho}_\kappa(\mathbb{M}, \ell, w_{[i]}, \Delta_i, \beta_i)$ 输出一个 *ρ*-接受基 $\mathbb{M}'$，其中 *$\rho(\cdot)$* 查询的分布被 $\mathsf{D}^\ell_\kappa(\Delta_i, \beta_i)$ 所限制。

定义 *26*（递归提取器）。设 $\mathbb{M} = (\mathbb{M}_1, \ldots, \mathbb{M}_r)$ 是一个格树，假设 $\mathcal{R}$ 是一个针对形式为 $(\{w_{[i]}\}, \mathbb{M}^*_{i+1}, \ldots, \mathbb{M}^*_r)$ 的格树的 $(\epsilon_i, \mathsf{D}^\ell_i(\Delta,\beta))$-接受基提取器。递归提取器 $\mathcal{R}_\kappa[\mathcal{R}]$，其中 $\kappa = (\alpha_i, \lambda_i)$ 和 $\alpha_i\ \lambda_i \in (0,1)$，在输入 $(\mathbb{M}, \ell, w_{[i-1]}, \Delta_{i-1}, \beta_{i-1})$ 上按如下方式工作。

1. 设置 $\Delta_i = \alpha_i(\Delta_{i-1} - \ell\omega_{\mathbb{M}_i})$，$\beta_i = \frac{1}{3}(1-\alpha_i)$，$k = k^{\mathsf{D}_i}(\lambda_i)$，和 $\mu = \mu_{\mathsf{D}^\ell_i(\Delta_i,\beta_i)}$.
2. 设置 $B^* = \varnothing$ 和 $T = \varnothing$.
3. 当 $|B^*| < d_i$:，
   (a) 计算 $w = \mathcal{S}^{f\ \rho}_{\alpha_i}(\mathbb{M}, \ell, w_{[i-1]}, B^*, \Delta_{i-1}, \beta_{i-1})$，
   (b) 提取子树 $t = \mathcal{R}^{f\ \rho}(\mathbb{M}, \ell, w_{[i]}, \Delta_i, \beta_i)$，，但中断执行并，
   如果它试图进行超过 $k\mu$ 次查询，则设置 $t = \perp$.
   (c) 如果 $t \neq \perp$，则设置 $B^* = B^* \cup \{w_i\}$ 和 $T = T \cup \{t\}$。
4. 返回接受基树 T。

以下引理和推论源自于 [12]，中相应的证明，我们在其中使用索引来说明与原始递归公式的相似性，且有所变通。

引理 *5 (*递归提取器*)*。算法 $\mathcal{R}_\kappa[\mathcal{R}]$ *is a* $(\epsilon_{i-1}, \mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1}))$- 接受基提取器，其中 $\epsilon_{i-1} = \epsilon_i/\alpha_i + \ell\omega_{\mathbb{M}_i}$ 和

$$\mathcal{G}_{\mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1})}(z) = \prod_{i=1}^{d_i} \mathcal{G}_{\mathsf{Geo}(\beta_i\lambda_i)}\big(\mathcal{G}_{\mathsf{Geo}(\beta^2_{i-1}\Delta_{i-1}/\ell)}(z) z^{k^{\mathsf{D}_i}(\lambda_i)\mu_{\mathsf{D}^\ell_i(\Delta_i,\alpha_i)}}\big) \ , \quad (19)$$

由 $\beta_i = \frac{1}{3}(1-\alpha_i)$ 和 $\Delta_i = \alpha_i(\Delta_{i-1} - \ell\omega_{\mathbb{M}_i})$ 定义。

推论 1 (递归提取器)。分布 $\mathsf{D}^\ell_{i-1}(\Delta,\beta)$ 满足

$$\mu_{\mathsf{D}^\ell_{i-1}(\Delta_{i-1},\beta_{i-1})} = \frac{3d_i}{(1-\alpha_i)\lambda_i}\left(\frac{\ell}{\beta^2_{i-1}\Delta_i} + k^{\mathsf{D}_i}(\lambda_i)\mu_{\mathsf{D}^\ell_i(\Delta_i,\alpha_i)}\right) \quad (20)$$

$$t^{\mathsf{D}_{i-1}}_{d_i}(k) \le t^{\mathsf{CG}}_{d_i}(k) \quad \textit{for } k \in (1,\infty) \ . \quad (21)$$

如果 $\beta^2_{i-1} \ge \beta^2_i\alpha_i$，那么可以通过观察到初始样本可以在递归调用中重复使用，并且递归调用稍微更昂贵来丢弃术语 $\ell/(\beta^2_{i-1}\Delta_i)$。当 $\beta_i = \frac{1}{3}(1-\alpha_i)$ 总是这样时，因为 $(1-\bar{\alpha}_{i-1})^2 \ge \alpha_i(1-\alpha_i)^2$ 对于 $\alpha_{i-1}\ \alpha_i \in (0,1)$，。

因此，与基本情况相比，在 [12] 中期望值的变化仅在每个递归调用中乘以一个因数三（并且有 $r-1$ 层递归），并且基本提取器的期望值增加了 $3^2\ell/(1-\alpha_{r-1})^2$ 倍。设置 $\nu_i = 1/\alpha_i$ 即得该定理。

D Grafting Prover

An honest grafting prover obviously does not need the liberty to graft additional branches to an execution.

Definition 27 (Grafting Prover). *If $(\mathcal{P}, \mathcal{V})$ is a $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$-special sound protocol, then on common input x, and private input w such that $(x, w) \in \mathsf{R}$, the* grafting prover $\mathsf{G}[\mathcal{P}]$ *proceeds as follows.*

1. *Start a simulation of $\mathcal{P}$ on input (x, w) and when it outputs a message a_0, hand $(0, a_0)$ to the verifier.*
2. *For $i = 1, \ldots, r$:*
   (a) *Wait for a message v_i from the verifier.*
   (b) *Continue the simulation of $\mathcal{P}$ on input v_i, until it outputs a message a_i, hand (i, a_i) to the verifier.*

E Omitted Proofs

Proof (Lemma 1). We have $\mathrm{E}\left[1/\delta_X \,|E\right] = \sum_{x\in[X]} \Pr\left[X = x \,|E\right]/\delta_X = 1/\Pr\left[E\right] \leq 1/\Delta$. Markov's inequality then implies $\Pr\left[\delta_X < \alpha\Delta \,|E\right] = \Pr\left[1/\delta_X > 1/(\alpha\Delta) \,|E\right] \leq \alpha$.

F Basic Definitions

Definition 28 (Matroid). *A* matroid *is a pair (S, I) of a* ground set *S and a set $I \subset 2^S$ of* independence sets *such that:*

1. *I is non-empty,*
2. *if $A \in I$ and $B \subset A$, then $B \in I$, and*
3. *if $A, B \in I$ and $|A| > |B|$, then there exists an element $a \in A \setminus B$ such that $\{a\} \cup B \in I$.*

Definition 29 (Submatroid). *Let (S, I) be a matroid and $S' \subset S$. The* submatroid *induced by S' is the pair (S', I') defined by $I' = I \cap 2^{S'}$.*

Definition 30 (Basis). *Let (S, I) be a matroid. A set $B \in I$ such that $B \cup \{x\} \notin I$ for every $x \in S \setminus B$ is a* basis.

Definition 31 (Rank). *The* rank *of a matroid (S, I) is the unique cardinality of each basis in I.*

Definition 32 (Rank of Set). *Let (S, I) be a matroid and $A \subset S$. The* rank $\mathsf{rank}(A)$ *of A is the rank of the submatroid induced by A.*

Definition 33 (Span and Flats). *Let (S, I) be a matroid and $A \subset S$. The* span *of A is defined by $\mathsf{span}(A) = \{x \in S \mid \mathsf{rank}(A \cup \{x\}) = \mathsf{rank}(A)\}$ and A is a* flat *if $\mathsf{span}(A) = A$.*

D 接枝证明器

一个诚实的 grafting 证明者显然不需要自由来将额外的分支 graft 到执行上。

定义 *27* (接枝证明器)。如果 *(P, V)* 是一个 $(\mathbb{M}_1, \ldots, \mathbb{M}_r)$-特殊声音协议，那么在公共输入 x 和私有输入 w 满足 $(x, w) \in \mathsf{R}$ 的情况下，接枝证明器 *G*[*P*] 按以下方式执行。

1. 在输入 *(x, w)* 上开始 $\mathcal{P}$ 的模拟，当它输出消息 a_0 时，将 *(0, P)* 交给验证器。*(0,* a_0*)* 交给验证器。
2. 对于 $i = 1, \ldots, r$:
   (a) 等待验证者发送消息 v_i 。
   (b) 在输入 v_i上继续模拟 $\mathcal{P}$ ，直到它输出消息 a_i，将 *(i,* a_i*)* 交给验证者。

E 省略的证明

证明（引理1）。我们有 $\mathrm{E}\left[1/\delta_X \,|E\right] = \sum_{x\in[X]} \Pr\left[X = x \,|E\right]/\delta_X = 1/\Pr\left[E\right] \leq 1/\Delta$。马尔可夫不等式则意味着 $\Pr\left[\delta_X < \alpha\Delta \,|E\right] = \Pr\left[1/\delta_X > 1/(\alpha\Delta) \,|E\right] \leq \alpha$。

F 基本定义

定义*28*（格）。格是一个有序对 *(S, I)*，包含一个基础集 S 和一个独立集的集合 $I \subset 2^S$，满足:

1. I 非空，
2. 若 $A \in I$ 和 $B \subset A$，则 $B \in I$，且
3. 若 A、$B \in I$ 和 $|A| > |B|$，则存在一个元素 $a \in A \setminus B$，使得 $\{a\} \cup B \in I$。

定义 *29*（子格）。设 *(S, I)* 是一个格， $S' \subset S$。由 S' 诱导的子格是定义为 $I' = I \cap 2^{S'}$ 的对 *(S′, I′)*。

定义 *30*（基）。设 *(S, I)* 是一个格。一个集合 $B \in I$，使得对每个 $x \in S \setminus$，B 是一个基。

定义*31*（秩）。矩阵（*S, I*）的秩是每个基的独特色数，在 *I* 中。

定义*32*（集合的秩）。设（*S, I*）是一个矩阵， $A \subset S$。*A*的秩*rank(A)*是*A* 诱导的子矩阵的秩。

定义*33*（扩张和平面）。设（*S, I*）是一个矩阵， $A \subset S$。 *A*的扩张由*span* $(A) = \{x \in S \mid$ *rank(*$A \cup \{x\}$*) = rank(A)*$\}$ 定义，且 A 是一个平面当且仅当 $\mathsf{span}(A) = A$。